Quizrr

India's Most Comprehensive & the Most Relevant Test Series designed according to the latest pattern of exams!

JEE MAIN | JEE ADV. | BITSAT | WBJEE | MHT CET

and many more...

Click here to join Test Series for 2022

It's time for you to crack upcoming IIT JEE Main & Advanced and other competitive exams with India's Most Trusted Online Test Series. **Many questions at JEE Main 2021 were same/similar to the ones asked in our test series.** That's the power of our test series!

Trusted by thousands of students & their parents across the nation

Our result in JEE Main 2021

150+
Got **99+ percentile** (overall)

301
Got **99+ percentile** in one or more subjects

85%
Improved their score by **25 percentile**

89%
Felt **overall confident** after the test series

Click here to join Test Series for 2022

MARKS

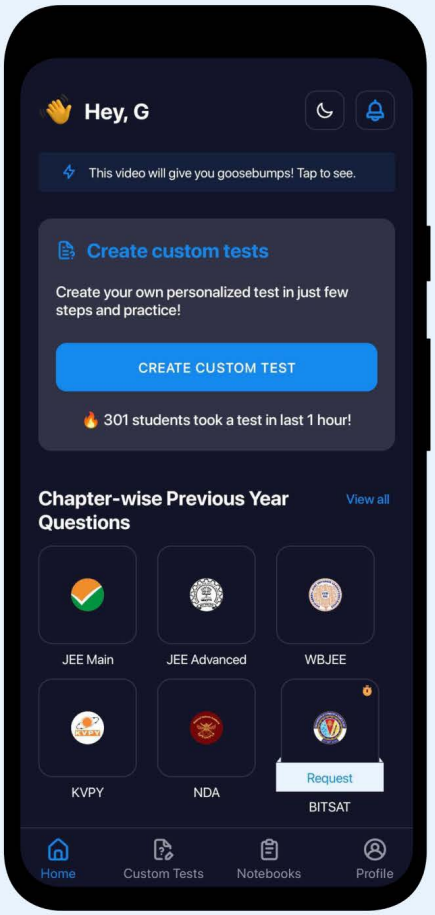
FREE Question Bank & Previous Year Questions for

JEE MAIN | JEE ADV. | BITSAT | WBJEE | MHT CET

and many more...

GET IT ON Google Play

Click on this button



😎 Why download MARKS?

- 📚 Chapter-wise PYQ of JEE Main, JEE Advanced, NEET, AIIMS, BITSAT, WBJEE, MHT CET etc.
- 📚 Chapter-wise NTA Abhyas questions
- 🎯 Daily practice challenge and goal completion
- 🔖 Bookmark important questions and add them to your notebooks
- ✍️ Create unlimited Custom Tests

And all this for FREE. Yes, FREE! So what are you waiting for, download MARKS now.

4.8	30,000+	1,00,000+
Rating on Google Play	Students using daily	Questions available

GET IT ON Google Play

Click on this button

THERMODYNAMICS

Quizrr

THERMODYNAMICS

Some Basic Terms

(a) **Thermodynamics :** The branch of science which deals with transformation of heat energy into other forms of energy or vice-versa.

(b) **Thermal equilibrium :** If two systems are at same temperature, they are said to be in thermal equilibrium.

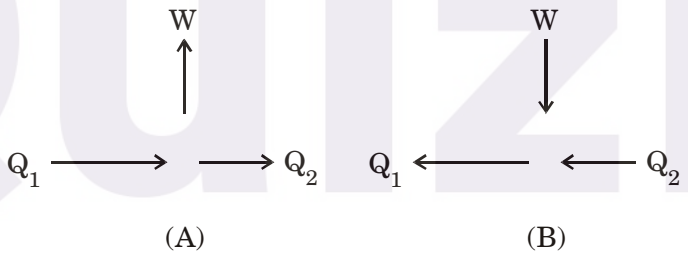
(c) **Thermodynamical equilibrium :** A system is said to be in thermodynamical equilibrium when it is in a state of thermal, mechanical and chemical equilibrium.

(d) **Thermodynamical variables :** These are macroscopic physical quantities like pressure (P), volume (V) and temperature (T), etc., which are used to describe the state of the system. The relation between these variables is called equation of state.

(e) **Thermodynamical process :** If the state of a system changes in such a way that any of P, V, T, etc. changes, the process is called thermodynamical process.

(f) **Cyclic process :** If a system after undergoing through a series of changes comes back to its initial state, the process is called cyclic.

(g) **Reversible process :** A reversible process is one which can be reversed in such a way that all changes taking place in the direct process are exactly replaced in the inverse order and opposite sense and no changes are left behind.



Reversibility in physics is an ideal concept that cannot be realised in practice.

I. Zeroth Law

According to it, "If systems A and B are each in thermal equilibrium with a third system C, then A and B are in thermal equilibrium with each other".

Regarding zeroth law it is worth noting that :

It defines temperature as the property of the system which equals that of another system when the two systems are in thermal equilibrium, e.g., if two gases in different containers at different pressures, temperature and volumes are put in thermal contact for sufficient time to reach thermal equilibrium, their pressures will in general not be equal, nor will their volumes; their temperatures, however, will always be equal in thermal equilibrium.

II. First Law of Thermodynamics

First law is a relation among heat, work and internal energy; so before dealing with the law here we review these physical quantities.

(a) Heat (ΔQ)

It is the energy that is transferred between a system and its environment because of temperature difference between them.

In case of solids and liquids,

$\Delta Q = mL$ if there is change of state

$\Delta Q = mC\Delta T$ if there is change in temperature

In case of gases,

$Q = nC\Delta T$

in differential form $dQ = nCdT$

where C is the molar heat capacity of the gas and n is the number of moles of the gas.

and always take, $\Delta T = T_f - T_i$

where T_f is the final temperature and T_i is the initial temperature of gas. Also, the molar heat capacity of an ideal gas

$$\boxed{C = \frac{R}{\gamma - 1} + \frac{R}{1 - x} = C_V + \frac{R}{1 - x}}$$

$C = C_V = \dfrac{R}{\gamma - 1}$ in isochoric process and

$C = C_P = C_V + R$ in isobaric process.

Note :

(1) SI unit of heat is Joule and CGS is calorie [1 cal = 4.2 J]

(2) It is path dependent, e.g., heat required to change the temperature of same gas through same amount at constant volume.

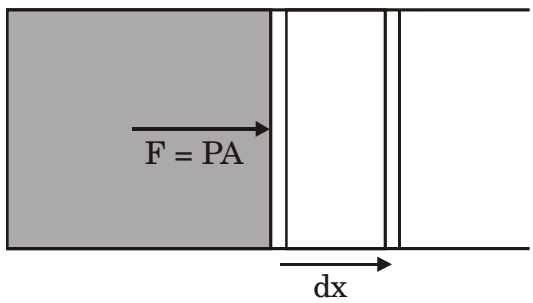
(b) Work (ΔW)

Work can be defined as the energy that is transferred from one body to other owing to a force that acts between them. The amount of work done by a system as it expands or contracts is given by

$$\Delta W = \int dW = \int_{v_i}^{v_f} PdV$$

Derivation

Figure shows a gas confined to a cylinder that has a movable piston at one end. If the gas expands against the piston, it exerts a force through a distance and does work on the piston. If the piston compression the gas as it is moved inward, work is also done–in this case on the gas. The work associated with such volume changes can be determined as follows.



Let the gas pressure on the piston face be P. Then the force on the piston due to gas is PA, where A is the area of the face.

When the piston is pushed outward an infinitesimal distance dx, the word done by the gas is

$$dW = F \cdot dx = PA\ dx$$

which, since the change in volume of the gas is dV = Adx, becomes

$$dW = PdV$$

For a finite change in volume from V_i to V_f, this equation is then integrated between V_i to V_f to find the net work

$$W = \int dW = \int_{V_i}^{V_f} PdV$$

Note :

(1) Like heat, work is also path dependent, scalar physical quantity.

(2) In a cyclic process work done is equal to the area under the cycle and is positive if the cycle is clockwise and negative if anticlockwise.

(c) Internal Energy (ΔU) :

Internal energy of a system is the energy possessed by the system due to molecular motion and molecular configuration.

$$\Delta U = n\ C_V dT$$

This equation can be applied in any process, whether it is isobaric, isothermal, adiabatic or else.

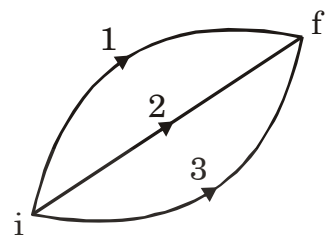
Note :

(1) Change in internal energy is path independent and depends only on the initial and final states of the system.

(2) Change in internal energy in a cyclic process is always zero $\Delta U = U_F - U_C = 0$

The Law

We now come to the first law.



Let a system changes from an initial equilibrium state i to a final equilibrium state f in a definite way, the heat absorbed by the system being Q and the word done by the system being W. Then we compute the Q – W. While Q and W both depend on the thermodynamic path taken between two equilibrium states, their difference Q – W does not. We do this over and over again, using different paths each time. We find that in every case the quantity Q – W is the same. In thermodynamics there is a function of the thermodynamic coordinates (P, V and T) whose final value minus its initial value equals the change Q – W in the process. We call this function the internal energy function. We have,

$$\boxed{\Delta U = U_f - U_i = Q - W} \qquad ...(i)$$

This equation is known as the first law of thermodynamics.

We describe the energy relations in any thermodynamics process in terms of the quantity of heat Q added to the system and the work W done by the system. Both Q and W may be positive, negative or zero. A positive value of Q represents heat flow heat flow into the system, negative Q represents heat flow out of the system. A positive value of W represents work done by the system against its surroundings, such as work done by an expanding gas. Negative W represents work done on the system by its surroundings such as work done during compression of a gas.

Thermodynamic sign conventions for heat and work

Process	Conventions
Head added to the system	$Q > 0$
Heat removed from the system	$Q < 0$
Work done by the system	$W > 0$
Work done on the system	$W < 0$

Equation (i) can be written as

$$Q - W_1 = Q_2 - W_2 = ...$$

or $$\Delta U_1 = \Delta U_2 = ...$$

that is the change in the internal energy of the system between two points is path independent. It depends on thermodynamic coordinates of the two points. For example, in case of an ideal gas it depends only on the initial and final temperatures.

Often the first law must be used in its differential form, which is

$$dU = dQ - dW \quad ...(ii)$$

This can also be written as

$$dQ = dU + dW \quad ...(iii)$$

or $$Q = \Delta U + W \quad ...(iv)$$

Note :

(1) For solids/liquids pressure is assumed to be constant through the process, $\Delta W = P_{atm}\,\Delta V$

(2) For gases, we will be frequently using the following results for calculations.

$$\Delta W = \int P dv$$

$$PV = nRT$$

Process equation

Example 1

A solid cylinder of mass 1 kg is initially kept at 20 C under atmospheric pressure, P_{atm} = 10^5 N/m^2. Now, a heat of 20000 J is given to the cylinder, then find :

(a) increase in temp. of cylinder

(b) work done by cylinder

(c) increase in its internal energy

S = 400 J/g/K, γ = 9 10^{-5} C^{-1}, d = 9000/g / m^3

Solution :

(a) $\Delta Q = ms\Delta\theta$

$$\therefore \quad \Delta\theta = \frac{\Delta Q}{ms} = \frac{20000}{1 \times 400} = 50°C$$

$\therefore$ final temp. of cylinder = 20 C + 50 C = 70 C

(b) $\Delta V = \gamma V \Delta\theta$

$$= \left(9 \times 10^{-5}\right)\left(\frac{1}{9000}\right) \times 50$$

$$= (5 \quad 10^{-7})$$

$\therefore$ Work done $= P_{atm}\ \Delta V$

$$= (10^5 \quad 5 \quad 10^{-7}) = (5 \quad 10^{-2})\text{J}$$

(c) increase in internal energy

$\Delta Q = \Delta U + \Delta W$

$\Delta U = \Delta Q - \Delta W = (20000 - 0.05)$

$\Delta U = 19999.95$ J

Example 2

1 kg of water initially at 10 C is given heat and then converted into steam, then find the increase in the internal energy.

$L_V = 2.5 \quad 10^6$ J/kg

$\rho_{steam} = 0.5$ kg/m^3

Solution :

$\Delta Q = ms\ \Delta\theta + mL_V$

$= 1 \quad 4200 \quad 90 + 1 \quad 2.5 \quad 10^6$ J

ΔV = vol. of steam – vol. of water

$$= \frac{1}{0.5} - \frac{1}{10^3} \simeq 2\text{m}^3$$

$\Delta W = P_{atm}\ \Delta V = (10^5 \quad 2)$J

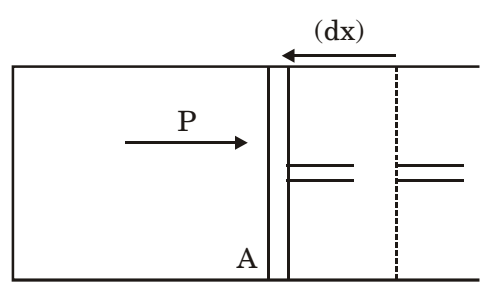
$\Delta U = (2.878 \quad 10^6 - 2 \quad 10^5) = (26.78) \quad 10^5$ J

For gaseous system

Work done by a gas

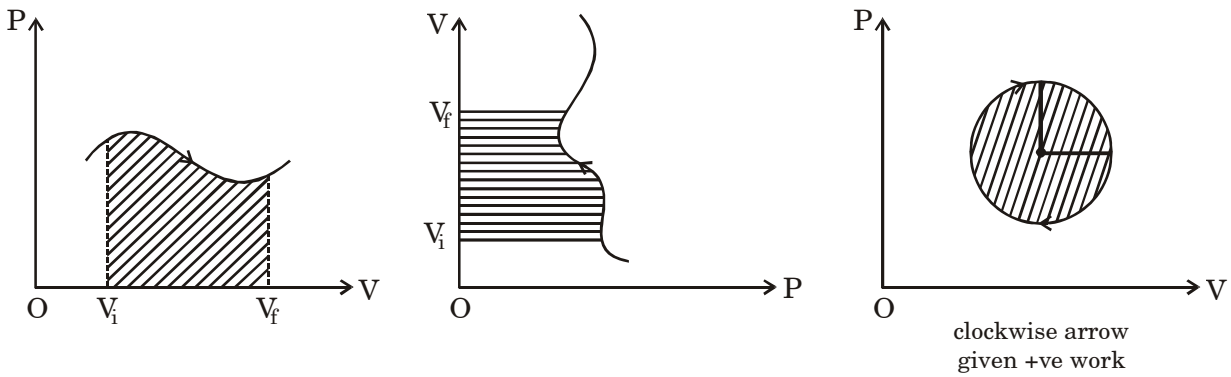
$dW = F\ .\ dx$

$dW = PA\ dx$



$$\boxed{\Delta W = \int_{v_i}^{v_f} PdV}$$

$\boxed{\Delta W = \text{area under the P-V curve.}}$



Work done in each case

= area under the P-V curve

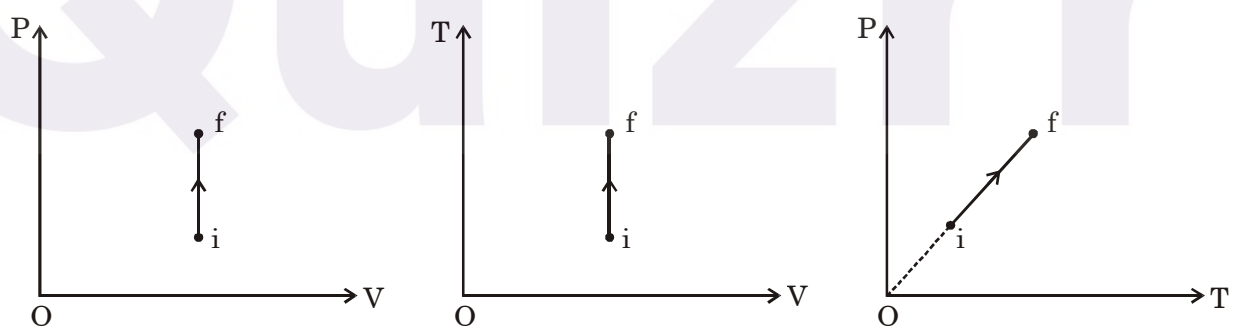
Different Thermodynamic Process

(1) **Isochoric Process :** The process during which the volume of the gaseous system remains constant, throughout entire process, is called isochoric process.

$$\boxed{V = \text{const}}$$

(i) $\Delta W = \int PdV = 0$

(ii) $\therefore$ $\boxed{\Delta Q = \Delta U}$



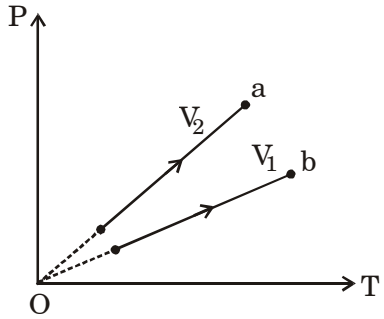
In the P-T graph [$\therefore$ PV = nRT]

$\therefore$ slope $\propto \dfrac{1}{V}$, slope $\propto$ n

Slope = $\left(\dfrac{nR}{V}\right)$ [$\therefore$ PV = nRT]

slope of a > slope of b

$\therefore$ $(V_2 < V_1)$



Example 3

2 mol of an ideal gas is taken through a process $PT^2 = K$. Find the work done by the gas from v_0 to $2v_0$ volume expansion, if initial pressure = P_0.

Solution :

$PT^2 = K \qquad PV = nRT$

$$P\left(\frac{PV}{nR}\right)^2 = K$$

$$P^3 V^2 = n^2R^2K$$

$$P = \left(\frac{n^2R^2K}{V^2}\right)^{1/3}$$

$$\therefore \quad W = \int_{V_0}^{2V_0} pdv$$

$$= \int_{V_0}^{2V_0} \left(n^2R^2K\right)^{1/3} V^{-2/3}\, dV$$

$$= \left(n^2R^2K\right)^{1/3} \int_{V_0}^{2V_0} V^{-2/3}\, dV$$

$$= 3K \left|V^{1/3}\right|_{V_0}^{2V_0} = 3K\left[\left(2V_0\right)^{1/3} - \left(V_0\right)^{1/3}\right]$$

$$= 3K\, V_0^{1/3}\left[\sqrt{2} - 1\right]$$

---

(2) Isobaric Process : The process during which the pressure of the gaseous system remains constant throughout the entire process.

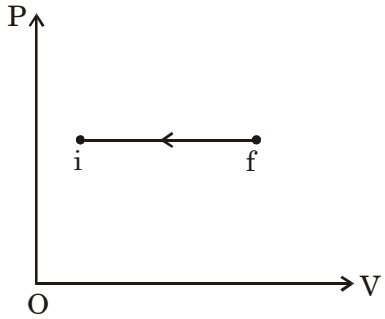
$$\boxed{P = \text{const.}}$$

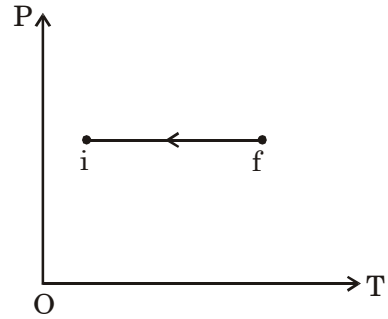
(i) $\Delta W = \int PdV = P\int dV = P\Delta V = (nR\Delta T)$

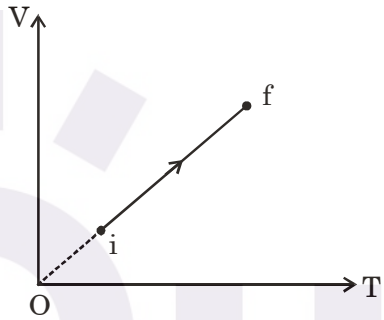
$(\Delta W = P\Delta V) = nR\Delta T)$

$$\begin{bmatrix} PdV + VdP = nRdT \\ \because\ dW = PdV = nRdT \\ \Delta W = P\Delta V = nR\Delta T \end{bmatrix}$$

(ii)

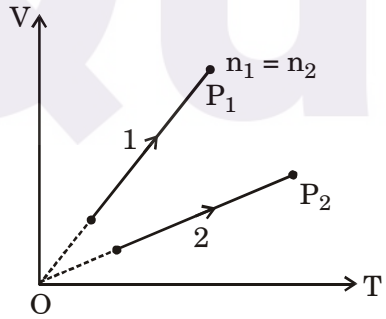




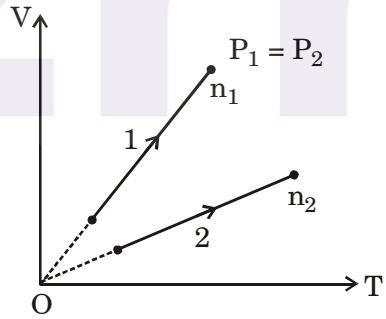


$$\text{slope} = \left(\frac{nR}{P}\right)$$

$$\therefore\ \text{slope}\ \alpha \left(\frac{n}{P}\right)$$



$(P_1 < P_2)$



$(n_1 > n_2)$

(3) Isothermal Process :

The process during which the temp. of the gaseous system remains constant during the entire process.

$$\boxed{T = \text{const}}$$

(i) Process equation = (PV = K)

(ii) Work done is an isothermal process

$$W = \int PdV = \int \frac{K}{V}\, dV = K\left[\ln V\right]_{V_i}^{V_f}$$

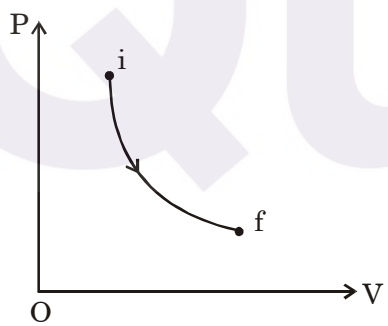
$$\therefore \quad W = nRT \log_e\left(\frac{V_f}{V_i}\right)$$

$$W = nRT \ln\left(\frac{P_i}{P_f}\right) \qquad \{ P_iV_i = P_fV_f = K \}$$

$$W = P_iV_i \ln\left(\frac{P_i}{P_f}\right)$$

(iii) $\Delta U = 0$ $\quad [\therefore T = \text{const}]$

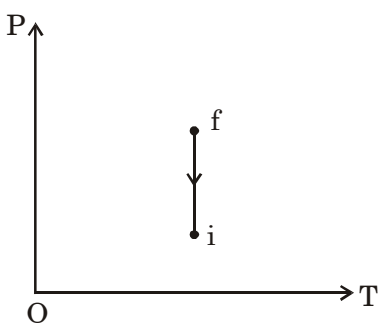
$$\boxed{\Delta Q = W}$$

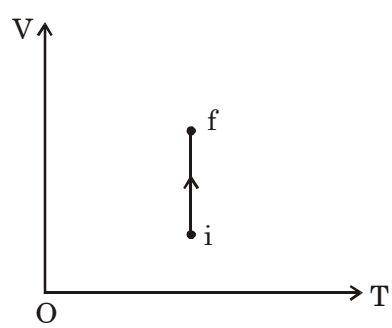


$PV = K$

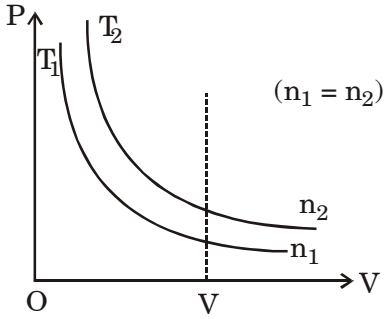
$$P = \left(\frac{K}{V}\right) \qquad \therefore \quad \left(\frac{dP}{dV}\right) = (-1)\, KV^{-2} < O$$

$$\therefore \quad \left(\frac{d^2P}{dV^2}\right) = 2KV^{-3} > O$$





Comparison between two P-V plot for an isothermal process



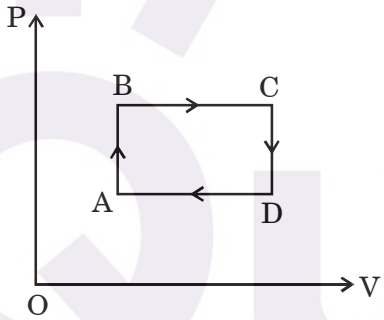
$P_2 > P_1$ for V

$\therefore$ $T_2 > T_1$

Cyclic Process

The process during which the initial and the final states of the gaseous system are same, is called cyclic process.

→ Cyclic process is represented as a closed loop on the P-V diagram.



$(T_i = T_f)$

$(u_i = u_f)$

$\Delta U = (u_f - u_i) = 0$

$\Delta Q_{cyclic} = \Delta W_{cyclic}$

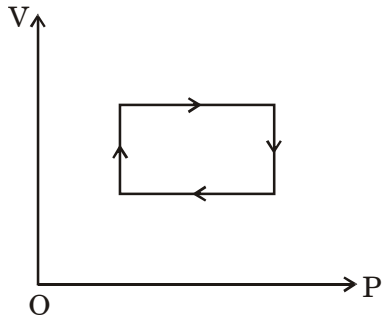
Work done in a cyclic process

= area of the loop in the P-V diagram

$\Delta W > 0$, if PV loop is clockwise

$\Delta W < 0$, if PV loop is counter clockwise

Note : In the V-P diagram, signs of work is reversed.



CW $\rightarrow \Delta W < 0$

CCW $\rightarrow \Delta W > 0$

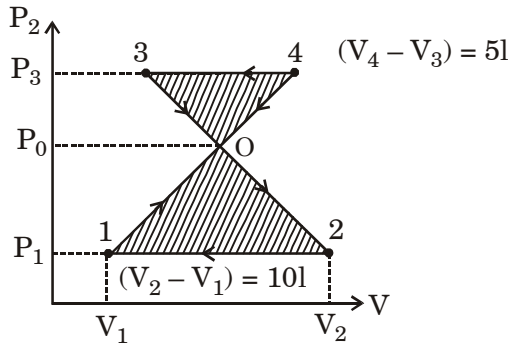
Example 4

Find the work done by an ideal gas during cyclic process (1432) as shown :

$P_3 = 4 \quad 10^5$ Pa

$P_0 = 3 \quad 10^5$ Pa

$P_1 = 10^5$ Pa



Solution :

Area of Δ (102) $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 2 \times 10^5 \times 10 \times 10^{-3}$ (CW)

$= (+\ 10^3\ J)$

Area of Δ (304) $= \frac{1}{2} \quad b \quad h$

$= \frac{1}{2} \quad (5 \quad 10^{-3}) \quad 10^5$ [Obtain, base length from similar triangle concept]

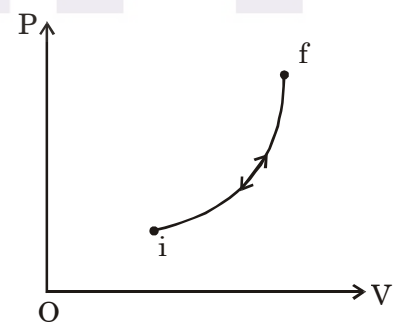
$= -\ 2.5 \quad 10^2 \equiv -\ (250\ J)$ CCW

$= (-\ 250\ J)$

$\therefore$ Work done during the entire 1432 prove

$= +\ 1000 + (-\ 250)\ J = 750\ J.$

Reversible Process



A process is said to be reversible, if, it is :

(i) quasi-static

(ii) Non-dissipative

(i) **Quasi static :** If unique value of P, V and T can be specified to a gaseous system at certain moment, then system is said to be in thermodynamic equilibrium.

But, in practical, no system can exactly achieve thermodynamic equilibrium because the parts of the system do not at once achieve, the same value of P, V, T.

$\therefore$ If the process is carried out very slowly then system will be in almost thermodynamic equilibrium and this process is called quasi-static process.

(ii) **Non-dissipative**

If no heat dissipation takes during the process then it is called non-dissipative.

Relation between C_P and C_V

$$C_P = \frac{(dQ)}{ndT}P, \qquad C_V = \frac{(dQ)}{ndT}V$$

$(dQ)_P = nC_PdT;$ $\qquad (dQ)_V = nC_VdT$

From Ist law of Thermodynamics

$(dQ)_V = (dU) + dW$

$nC_Vdt = dU + 0$

$\therefore \quad \boxed{du = nC_V\ dT}$

At const. pressure

$(dQ)_P = dU + dW$

$\Rightarrow \quad nC_P\ dT = nC_V\ dT + PdV$

$\Rightarrow \quad nC_PdT = nC_V\ dT + nRdT$

$\therefore \quad C_P = C_V + R$

$\therefore \quad \boxed{(C_P - C_V) = R} \qquad \left(\frac{C_P}{C_V}\right) = \gamma$

$\therefore \quad \boxed{C_V = \frac{R}{\gamma - 1}} \quad \boxed{C_P = \frac{\gamma R}{\gamma - 1}}$

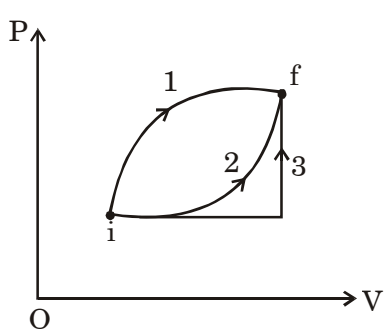
Note : $C_P - C_V = R/J$ when C_P and C_V are is cal and J is mechanical heat equivalent
$J = 4.18$ J/cal

Remember :

$\boxed{dU = nC_VdT}$ (Applicable for all process)

Example 5

In all the 1, 2, 3 process compare C_1, C_2 and C_3.



Solution :

From the diagram its, obvious that

$W_1 > W_2 > W_3$

$\Delta U_1 = \Delta U_2 = \Delta U_3$ (final, initial state)

$\therefore \quad \Delta Q_1 > \Delta Q_2 > \Delta Q_3$

$\therefore \quad \boxed{C_1 > C_2 > C_3}$

Adiabatic Process

The process during which is no heat exchange between the system and the surrounding for the entire process is called adiabatic process.

$$\boxed{dQ = 0}$$

[do not write it as $\Delta Q = 0$, This is because in a process, the net heat change may be zero but in an adiabatic process, there is no heat change any step of the process]

Derivation of Process equation

$dQ = 0$

$dQ = dU + dW$

$0 = nC_V dT + PdW$

$$n\left(\frac{R}{\gamma - 1}\right)dT + \frac{nRT}{V}dV = 0$$

$$-(\gamma - 1)\int \frac{dV}{V} = \int \frac{dT}{T}$$

$\ln T = -(\gamma - 1)\ln V + \ln K$

$\ln T + \ln V^{(\gamma - 1)} = \ln K$

$TV^{(\gamma-1)} = K$

$PV^{\gamma} = K$ (Use gas equation)

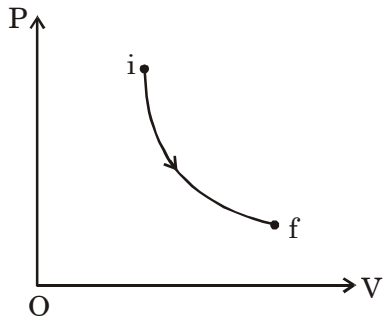
Process equation

$PV^{\gamma} = K$ ...(1)

$TV^{\gamma-1} = K$ ...(2)

$P^{1-\gamma}\ T^{\gamma} = K$ ...(3)

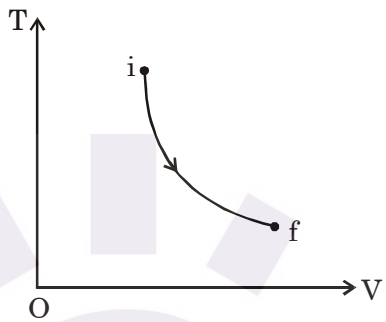
Plots in case of Adiabatic Process



$PV^{\gamma} = K$

$$\frac{dP}{dV} = -\gamma KV^{-\gamma-1} < 0$$

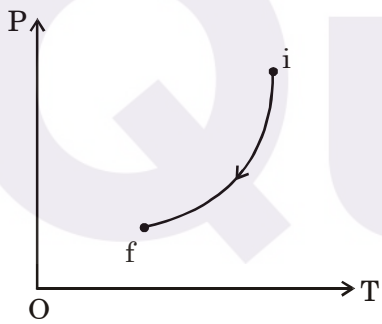
$$\frac{d^2P}{dV^2} = \gamma(\gamma+1)KV^{-\gamma-2} > 0$$



$TV^{\gamma-1} = K$

$$\frac{dT}{dV} = -(\gamma-1)KV^{-\gamma} < 0$$

$$\frac{d^2T}{dV^2} = +(\gamma-1)KV^{-\gamma-1} > 0$$



$P^{1-\gamma}T^{\gamma} = K$

$$P = K\ T^{\frac{\gamma-\gamma+1}{\gamma-1}}$$

$$\frac{dP}{dT} = \left(\frac{\gamma}{\gamma-1}\right)KT^{\gamma-\gamma+1} > 0$$

$$\frac{d^2P}{dT^2} = \left(\frac{\gamma}{\gamma-1}\right)\left(\frac{1}{\gamma-1}\right)KT > 0$$

Example 6

If $C_P = (a + KT)$ a = 27.68 J/mol/K

C_V = (b + KT) b = 19.37 J/mol/K

Derive the process eqn. for adiabatic process.

Solution :

$dQ = dU + dW$

$0 = n\ C_V\ dT + PdV$

$$nC_V dT + \frac{nRT}{V} dV = 0 \qquad n(b + KT)dT = -n\frac{RTdV}{V}$$

$$\therefore \quad \int_{T_1}^{T_2} \frac{(b + KT)}{T} dT = -R \int_{V_1}^{V_2} \frac{dV}{V}$$

Polytropic Process

Any process is a polytropic process, if it can be expressed as

$$\boxed{PV^x = \text{const.}} \qquad \text{where } x = \text{polytropic constant.}$$

(i) $x = \gamma$ for adiabatic process

$x = 1$ for isothermal

$x = 0$ for isobaric

$x = \infty$ for isochoric

Work done in a polytropic process

$$\Delta W = \int_{V_1}^{V_2} PdV = K \int_{V_1}^{V_2} V^{-\gamma}\, dV$$

$$\boxed{\Delta W = \frac{P_2V_2}{1 - x}} \qquad \text{for } (x \neq 1)$$

Molar Heat Capacity in a Polytropic Process

$dQ = dU + dW$

$nCdT = nC_V dT + PdV$

$$nCdT = nC_V dT + \frac{nRT}{V} dV$$

$$\boxed{C = C_V + \frac{RT}{V}\frac{dV}{dT}}$$

$$\boxed{C = C_V + \frac{PdV}{ndT}} \qquad \text{...(1)}$$

This eqn. to be used when we have to obtain following : (in any general process)

molar heat capacity → process eqn. process eqn. → molar heat capacity

Now, $PV^x = K$

$$V^x dP + (xV^{x-1}\, dV)\, P = 0$$

$$VdP + (xdV)\, P = 0$$

$$VdP = -\, x\, PdV$$

we have from gas eqn.

$$(PdV + VdP = nRdT)$$

$$PdV\,(1 - x) = nRdT$$

$$\Rightarrow \quad \frac{PdV}{ndT} = \left(\frac{R}{1-x}\right)$$

Substitute in (1)

$$C = C_V + \frac{R}{1-x}$$

∴ In any polytropic process, molar heat capacity

$$\boxed{C = C_V + \frac{R}{1-x}}$$

Example 7

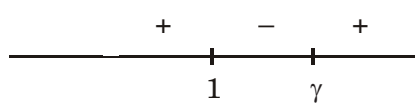
Find the values of polytropic const. x for which specific heat of the gas for the process is negative.

Solution :

$$C = C_V + \left(\frac{R}{1-x}\right)$$

$$= \frac{R}{(\gamma - 1)} + \frac{R}{(1-x)}$$

$$= \frac{R(1-x) - R(\gamma - 1)}{(\gamma - 1)(1-x)}$$



$$= \frac{R(\gamma - x)}{(\gamma - 1)(x - 1)}$$

C is –ve when $x \in (1, \gamma)$

$[1 > x < \gamma]$

Example 8

If $C = (C_V + 3at^2)$ then find the equation of process in terms of T and P.

Solution :

Use $C = C_V + \left(\frac{PdV}{ndT}\right)$

$$C_v + 3aT^2 = C_v + \frac{PdV}{ndT}$$

$$3aT^2 = \frac{PdV}{ndT}$$

$$3aT^2 = \left(\frac{RT}{V}\right)\frac{dV}{dT}$$

$$\left(\frac{3a}{R}\right)\int TdT = \int \frac{dV}{V}$$

$$\therefore \quad \ln V = \left(\frac{3a}{2R}\right)T^2$$

$$\therefore \quad V = e^{(3a/2R)T^2}$$

$$Ve^{-(3a/2R)T^2} = K$$

(ii) $C = C_V + \beta V$

Again, $C_V + \beta V = CV = \frac{PdV}{ndT}$

$$\beta V = \left(\frac{RT}{V}\frac{dV}{dT}\right)$$

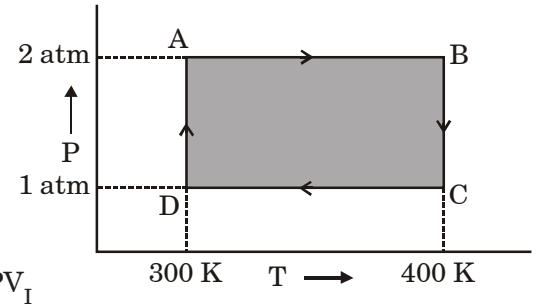
$$\Rightarrow \quad \frac{\beta}{R}\int \frac{dT}{T} = \int \frac{dV}{V^2}$$

$$\left(\frac{\beta}{R}\ln T\right) = \left(\frac{-1}{V}\right) + (\ln K)$$

Example 9

Two moles of helium gas undergo a cyclic process as shown. Assuming the gas to be ideal calculate the following quantities in the process : (a) the net work done (b) the net change in internal energy (c) the net change in heat energy. (R = 8.32 J/mol K)

Solution :



(a) As work done $\Delta W = \int PdV$

So at constant pressure,

$\Delta W = P[V_F - V_I] = nR[T_F - T_I]$

...[as for ideal gas $PV = nRT$; $PV_F = nRT_F$ and $PV_I = nRT_I$]

$$\Delta W = nRT \log_e \frac{V_F}{V_I} = nRT \log_e \frac{P_I}{P_F}$$

[as for constt. temperature, PV = constt., i.e., $P_F V_F = P_1 V_I$, i.e., $(V_F/V_I) = (P_I/P_F)$

So work done for path AB, BC, CD and DA respectively will be

$$\Delta W_{AB} = nR[T_F - T_I]$$
$$= 2 \quad R[400 - 300] = 200\ R$$
$$\Delta W_{BC} = nRT \log_e (P_I/P_F)$$
$$= 2 \quad R \quad 400 \log_e 2 = 800\ R \log_e 2$$
$$\Delta W_{CD} = nR[T_F/T_I]$$
$$= 2 \quad R\ [300 - 400] = -200R$$
$$\Delta W_{DA} = nRT \log_e (P_I/P_F)$$
$$= 2 \quad R \quad 300 \log_e (1/2)$$
$$= -600R \log_e 2$$

and hence, the work done in the complete cycle,

$$\Delta W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$
$$= 200R \log_e 2 = 200 \quad 8.32 \quad 0.693$$
$$= 1153\ J$$

(b) As this is a cyclic process,

$$U_F = U_I$$

so $\Delta U = U_F - U_I = 0$

(c) Now according to first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

So substituting the values ΔW and ΔU from parts (a) and (b) in the above,

$$\Delta Q = 0 + 1153 \text{ J} = 1153 \text{ J}$$ **Ans.**

Example 10

Three moles of an ideal gas [$C_P = (7/2)R$] at pressure P and temperature T is isothermally expanded to twice its initial volume. It is then compressed at constant pressure to its original volume. Finally the gas is compressed at constant volume to its original pressure (P). (a) Sketch P-V and P-T diagram for the process. (b) Calculate the work done by the gas and heat supplied to the gas during the complete cycle.

Solution :

(a) Initially the state A of the gas is described by (P, V, T). As for isothermal change T = constt. with PV = constt., so the state B will be [(P/2), 2V, T]. Now as for isobaric change P = constt. and $V \propto T$, so the state C will be [(P/2), V, (T/2)]. So the P-V and P-T curves will be as shown in Fig. (A) and (B) respectively.

(b) For isothermal change,

$$W_{iso} = nRT \log_e \frac{V_F}{V_I} = 3RT\log_e 2 = 2.08RT$$

For isobaric change,

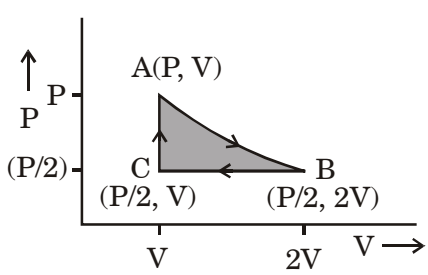
$$W_B = P(V_2 - V_1) = nR(T_F - T_I)$$

$$= 3R\left(\frac{T}{2} - T\right) = 1.5\,RT$$

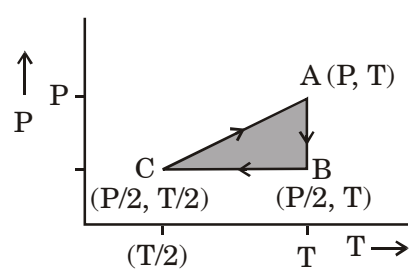
For isochoric change

$$W_C = 0$$

So $W_T = (2.08 - 1.5)RT = 0.58\ RT$ **Ans.**



(A)



(B)

Now as for cyclic process $U_F = U_I$, $\Delta U = 0$

$\therefore$ $\Delta Q = \Delta U + \Delta W = \Delta W = W_T = 0.58\ RT$ **Ans.**

Example 11

An ideal monatomic gas is confined in a cylinder by a spring loaded piston of cross-section 8×10^{-3} m^2. Initially the gas is at 300 K and occupies a volume of 2.4 10^{-3} m^3 and the spring is in its relaxed position. The gas is heated by a small electric heater until the piston moves out slowly by 0.1 m. Calculate (a) the final temperature of the gas and (b) the heat supplied (in J) by the heater. The force constant of the spring is 8000 N/m and atmospheric pressure 1 10^5 N/m^2.

Solution :

As here initially,

$P_I = P_0 = 1 \quad 10^5$ N/m^2, $V_I = V_0 = 2.4 \quad 10^{-3}$ m^3,

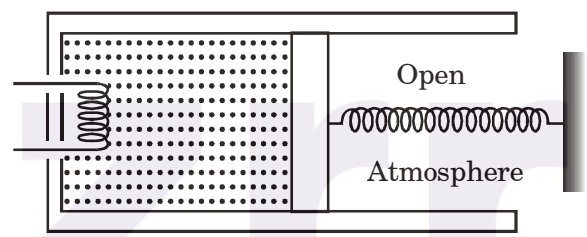
$T_I = T_0 = 300$ K

And finally, $$P_F = P_0 + \frac{Kx}{A} = 1 \times 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}}$$

$$= 2 \quad 10^5 \text{ N/m}^2$$

$$V_F = V_0 + Ax = 2.4 \quad 10^{-3} + 0.1 \quad 8 \quad 10^{-3}$$

$$= 3.2 \quad 10^{-3} \text{ m}^3$$



So from gas equation $PV = nRT$, i.e.,

$$\frac{P_1 V_I}{T_I} = \frac{P_F V_F}{T_F}$$

we have $$T_F = \frac{P_F}{P_I} \times \frac{V_F}{V_I} \times T_I$$

$$= \frac{2 \times 10^5}{1 \times 10^5} \times \frac{3.2 \times 10^{-3}}{2.4 \times 10^{-3}} \times 300$$

$$= 800 \text{ K} \qquad \textbf{Ans.}$$

(b) As $$\Delta W = \int P\, dV, \qquad \text{here} \quad P = P_0 + (kx/A)$$

So $$\Delta W = \int_0^{0.1} \left(P_0 + \frac{kx}{A}\right) A dx = \int_0^{0.1} (P_0 A + kx)\, dx$$

i.e., $$\Delta W = P_0 Ax + \frac{1}{2} kx^2$$

$$= [10^5 \quad 8 \quad 10^{-3} \quad 0.1 + \frac{1}{2} \quad 8000 \quad (0.1)^2]$$

i.e., $\Delta W = 80 + 40 = 120$ J

Note : 80 J of work is done against atmosphere and 40 J against spring.

and as $\Delta U = nC_V\Delta T = \dfrac{nR\Delta T}{(\gamma - 1)} = \dfrac{(P_F V_F - P_I V_I)}{(\gamma - 1)}$

$$\Delta U = \frac{[(2 \times 10^5) \times (3.2 \times 10^{-3}) - (1 \times 10^5) \times (2.4 \times 10^{-3})]}{[(5/3) - 1]}$$

$$\Delta U = \frac{3}{2} \times 10^2 [6.4 - 2.4] = 600 \text{ J}$$

Hence, total heat supplied,

$\Delta Q = \Delta U + \Delta W = 600 + 120 = 720$ J **Ans.**

Example 12

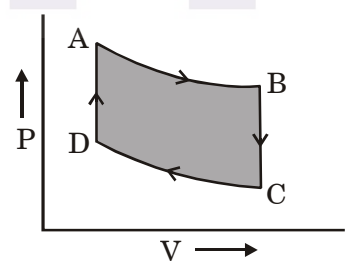
One mole of a monatomic ideal gas is taken through the cycle shown in Fig.

A → B adiabatic expansion

B → C cooling at constt. volume

C → D adiabatic compression

D → A heating at constt. volume



The pressure and temperature at A, B, etc., are denoted by P_A, T_A; P_B, T_B, etc. respectively. Given T_A = 1000K, $P_B = (2/3)P_A$ and $P_C = (1/3)P_A$. Calculate (a) the work done by the gas in the process A → B, (b) the heat lost by the gas in the process B → C and (c) temperature T_D. Given $(2/3)^{2/5} = 0.85$ and R = 8.31 J/mol K.

Solution :

(a) As for adiabatic change PV^γ = constt.,

i.e., $P\left(\dfrac{nRT}{P}\right)^r$ = constt. [as PV = nRT]

i.e. $\dfrac{T^\gamma}{P^{\gamma-1}}$ = constt.

So $\left(\frac{T_B}{T_A}\right) = \left(\frac{P_B}{P_A}\right)^{\gamma-1}$ with $\gamma = \frac{5}{3}$

i.e., $T_B = T_A\left(\frac{2}{3}\right)^{1-\frac{1}{\gamma}} = 1000\left(\frac{2}{3}\right)^{2/5} = 850\text{ K}$ ...(1)

So $W_A = \frac{nR[T_I - T_F]}{[\gamma - 1} = \frac{1 \times 8.31[1000 - 850]}{[(5/3) - 1]}$

i.e., $W_A = (3/2) \quad 8.31 \quad 150 = 1869.75\text{ J}$ **Ans.**

(b) For B $\rightarrow$ C, V = constt. so $\Delta W = 0$

So from first law of thermodynamics,

$\Delta Q = \Delta U + \Delta W = nC_V\Delta T + 0$

or $\Delta Q = 1 \times \left(\frac{3}{2}R\right)(T_C - 850)$ $\left[\text{as } C_V = \frac{3}{2}R\right]$

Now as along path BC, V = constt. $P \propto T$,

i.e., $\frac{P_C}{P_B} = \frac{T_C}{T_B}$,

i.e. $T_C = \frac{(1/3)P_A}{(2/3)P_A} \times T_B = \frac{T_B}{2} = \frac{850}{2} = 425\text{ K}$ ...(2)

So $\Delta Q = 1 \times \frac{3}{2} \times 8.31(425 - 850) = -5297.625\text{ J}$

Negative heat means heat is lost by the system. **Ans.**

(c) As A and D are on the same isochor.

$\frac{P_D}{P_A} = \frac{T_D}{T_A}$, i.e. $P_D = P_A\frac{T_D}{T_A}$

But C And D are on the same adiabatic,

$\left(\frac{T_D}{T_C}\right)^{\gamma} = \left(\frac{P_D}{P_C}\right)^{\gamma-1} = \left(\frac{P_A T_D}{P_C T_A}\right)^{\gamma-1}$

or $$(T_D)^{1/\gamma} = T_C\left[\frac{P_A}{P_C T_A}\right]^{1-\frac{1}{\gamma}}$$

i.e., $$T_D^{3/5} = \left(\frac{T_B}{2}\right)\left[\frac{P_A}{(1/3)P_A \times 1000}\right]^{2/5}$$

i.e. $$T_D^{3/5} = \left[\frac{1}{2}\left(\frac{2}{3}\right)^{2/5} \times 1000\right]\left[\frac{3}{1000}\right]^{2/5}$$

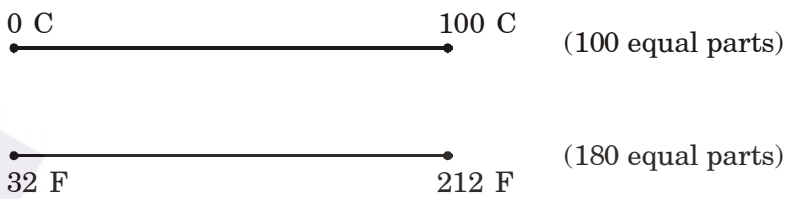
i.e. $T_D = 500$ K **Ans.**

TEMPERATURE SCALES

$T_C = T - 273.15$

Equation shows the relation between the temperatures in celsius scales and kelvin scale. Because the size of a degree is the same on the two scales, a temperature difference of 10 C is equal to a temperature difference of 10 K. The two scales differ only in the choice of the zero point. The ice point temperature on the kelvin scale, 273.15K, corresponds to 0.00 C and the Kelvin steam point 373.15 K, is equivalent to 100.00 C.

A common temperature scale in everyday use in US is the Fahrenheit scale. The ice point in this scale is 32 F and the steam point is 212 F. These two are divided in 180 equal parts. The relation between Celsius scale and Fahrenheit scale is derived below.

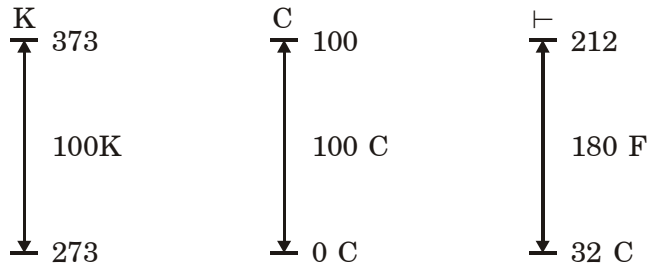


100 parts of Celsius scale = 180 parts of Fahrenheit scale

$$1 \text{ part of Celsius scale} = \frac{9}{5} \text{ parts of Fahrenheit scale}$$

Hence $$T_F = 32 + \frac{9}{5} T_C \quad \text{...(ii)}$$

Further $$\Delta T_C = \Delta T = \frac{5}{9} \Delta T_F \quad \text{...(iii)}$$



Relation among Kelvin, Celsius and Fahrenheit temperature scales

(I) Thermal Expansion :

With the increase in temp., due to increase in amplitude of vibrations of atoms (on molecules) the body expands on heating.

(1) Coefficient of linear expansion (α) :

The fractional charge is length for 1 C (or 1 K) change in temperature is called coefficient of linear expansion (α).

$$\boxed{\alpha = \left(\frac{\Delta l}{l\Delta\theta}\right)} \quad \text{(Average value)}$$

$\Delta l = l_0\alpha\Delta\theta$

$\therefore \quad l - l_0 = l_0\alpha\Delta\theta$

$$\boxed{l = l_0\,(1 + \alpha\Delta\theta)}$$

unit of α : C^{-1} or K^{-1}

Absolute Value :

$$\alpha = \lim_{\Delta\theta\to 0}\left(\frac{\Delta l}{l\Delta\theta}\right)$$

$$\therefore \quad \alpha = \left(\frac{dl}{l}\right)\left(\frac{1}{d\theta}\right)$$

$$\boxed{\int_{l_0}^{l}\frac{dl}{l} = \int_{\theta_0}^{\theta}\alpha d\theta}$$

$$\Rightarrow \quad \ln\left(\frac{l}{l_0}\right) = \int_{\theta_0}^{\theta}\alpha d\theta = a\Delta\theta \quad \text{(if } \alpha \text{ is constt.)}$$

$$\Rightarrow \quad \ln\left(\frac{1 + \Delta l}{l_0}\right) = \alpha\Delta\theta$$

$$\Rightarrow \quad \left(\frac{\Delta l}{l_0}\right) = \alpha\Delta\theta \quad \text{(approx.} \quad \therefore \ \ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots)$$

$$\therefore \quad \alpha = \left(\frac{\Delta l}{l_0\,\Delta\theta}\right) \quad \text{[Avg. value]}$$

Remember

(1) α, β, γ are defined only for solids.

(2) For fluids, only γ is defined.

(2) Coefficient of Superficial Expansion (β) :

The fractional change in area for 1 C (or 1 K) change in temperature is called coefficient of superficial expansion (β).

$$\boxed{\beta = \frac{\Delta A}{A\Delta\theta}} \text{(avg. value)}$$

$$\boxed{A = A_0 (1 + \beta\Delta\theta)}$$

Absolute value :

$$\beta = \lim_{\Delta\theta \to 0} \frac{\Delta A}{A\Delta\theta} = \left(\frac{dA}{A}\right)\left(\frac{1}{d\theta}\right)$$

$$\boxed{\int_{A_0}^{A} \frac{dA}{A} = \int_{\theta_0}^{\theta} \beta d\theta}$$

(3) Coefficient of Volume Expansion (γ) :

$$\gamma = \frac{\Delta V}{V\Delta\theta} \text{ (avg. value)}$$

The fractional change in volume for 1 C (or 1 K) change in temperature is called coefficient of volume expansion.

$V = V_0 (1 + \gamma\theta)$

Absolute value :

$$\gamma = \lim_{\Delta\theta \to 0} \left(\frac{\Delta V}{V\Delta\theta}\right) = \left(\frac{dV}{V}\right)\left(\frac{1}{d\theta}\right)$$

$$\therefore \quad \int_{V_0}^{V} \frac{dV}{V} = \int_{\theta_0}^{\theta} \gamma d\theta$$

Relation between α, β and γ :

$$\boxed{\alpha : \beta : \gamma = 1 : 2 : 3}$$

Derivation :

In a solid object having length, breadth and height l_0, b_0 and h_0, is being heated upto $\Delta\theta$.

$l = l_0(1 + \alpha\Delta\theta)$; $b = b_0 (1 + \alpha\Delta\theta)$; $h = h_0 (1 + \alpha\Delta\theta)$

$\therefore$ $A = lb = l_0b_0 (1 + \alpha\Delta\theta)^2 = l_0b_0 (1 + 2\alpha\Delta\theta) = l_0b_0 (1 + \beta\Delta\theta)$

[using binomial]

$\therefore$ $\beta = 2\alpha$

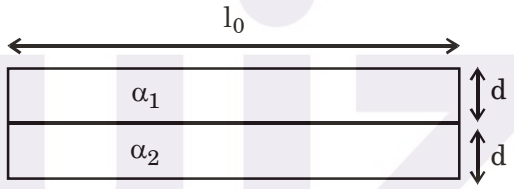
$V = lbh = l_0b_0l_2 (1 + \alpha\Delta\theta)^3 = l_0b_0h_0 (1 + 3\alpha\Delta\theta) = l_0b_0h_0 (1 + \gamma\Delta\theta)$

$\gamma = 3\alpha$

$\therefore$ $\alpha : \beta : \gamma = 1 : 2 : 3$

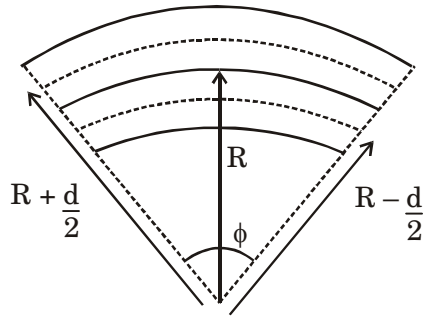
Effect of Temperature Change

(1) Bending of Bimetallic Strip



Let us consider a bimetallic strip of length l_0 and both metals having breadth d joint together. The 2 metals have coefficient of thermal expansion α_1 and α_2 respectively ($\alpha_1 > \alpha_2$)

Now, when the temperature of this strip is increased, then this strip has the characteristic property of bending due to unequal linear expansion of the two metals.

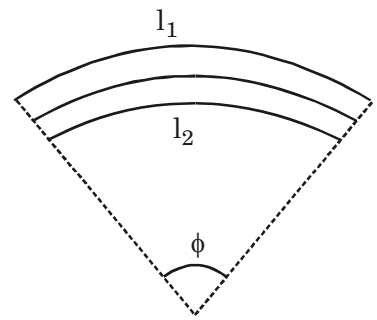


Bending will take place in such a manner that the one with higher coefficient (α) will be on convex side.

$$l_1 = l_0\ (1 + \alpha_1 \Delta\theta)$$
$$l_2 = l_0\ (1 + \alpha_2 \Delta\theta)$$

Since both subtend same angle at the centre,

$$\frac{l_1}{\left(R + \frac{d}{2}\right)} = \frac{l_2}{\left(R - \frac{d}{2}\right)}$$



$$\frac{(1 + \alpha_1\ \Delta\theta)}{(1 + \alpha_2\ \Delta\theta)} = \frac{R + d/2}{R - d/2}$$

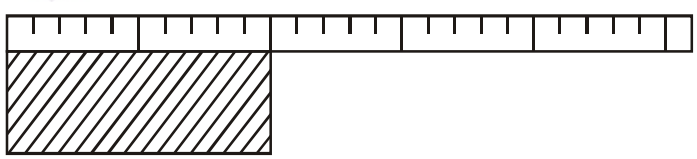
$$\Rightarrow \quad \frac{2 + \overbrace{(\alpha_1 + \alpha_2)}^{\text{neglected}}\Delta\theta}{(\alpha_1 - \alpha_2)\Delta\theta} = \left(\frac{2R}{d}\right)$$

$$= \quad \frac{2}{(\alpha_1 - \alpha_2)\Delta\theta} = \frac{2R}{d}$$

$$\boxed{R = \frac{d}{(\alpha_1 - \alpha_2)\theta}}$$

(2) Measurement from a heated scale (metallic)

If a scale gives correct reading at temperature θ, at temperature θ' ($> \theta$) due to linear expansion of scale, the scale will expand and scale reading will be lesser than true value. Similarly, when temperature is reduced, true value will be lesser than scale reading.



$$\boxed{\text{True value} = (\text{scale reading})\ (1 + \alpha\Delta\theta)}$$

– When the temperature is increased, reading of the scale decreases.

– When the temp. is decreased, reading from the scale is increased.

– When the object to be measured also expands on heating then use apparent α.

$$\boxed{\text{Apparent } \alpha = (\alpha_{scale} - \alpha_{body})}$$

$$TV = SR\ (1 + \alpha_{Apparent}\ \Delta\theta)$$

(3) Time gain/loss :

If a pendulum clock keeps proper time at temperature θ, at temperature θ′ (> θ) due to linear expansion, length of pendulum and hence time period will increase. Due to this increase in time period, in a given time interval (say t), the pendulum will exclude (t/T) oscillations instead of (t/T′) and so will lose [(t/T) – (t/T′)] oscillations. Now, as measurement of time is through the counting of oscillations, so in time t, a clock with time T′ will lose time by

$$\Delta t = T' \left[\frac{t}{T} - \frac{t}{T'}\right] = t\left[\frac{T'}{T} - 1\right]$$

also $$T = 2\pi\sqrt{\frac{l_0}{g}},\ T' = 2\pi\sqrt{\frac{l_0\,(1+\alpha\theta)}{g}}$$

$\therefore$ $$\frac{T^1}{T} = \sqrt{(1+\alpha\Delta\theta)}$$

$\therefore$ $$\Delta t = t\left(\sqrt{1+\alpha\Delta\theta} - 1\right)$$

Using Binomial expansion

$$\Delta t = t\left(1 + \frac{1}{2}\alpha\Delta\theta - 1\right) = \frac{1}{2}\alpha\, t\, \Delta\theta$$

$\therefore$ $$\frac{\Delta t}{t} = \frac{1}{2}\alpha\,\theta$$

or $$\boxed{\text{fractional loss/gain in time} = \frac{1}{2}\alpha\,\theta}$$

Example 13

Find the time loss in 1 year if temp of a pendulum clock is increased by 100 C ? ($\alpha = 10^{-5}\ C^{-1}$)

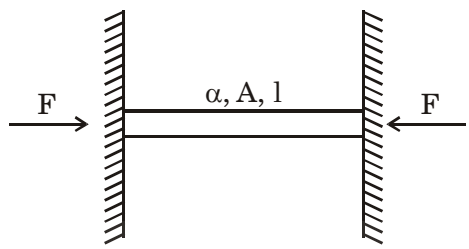
Solution :

Fractional time loss = $\left(\frac{1}{2}\alpha\theta\right)$

$\therefore$ time lost in 1 year, $\Delta t = \left(\frac{1}{2}\alpha\theta\right)$ 365 days

$= \left(\frac{1}{2} \times 10^{-5} \times 100 \times 365\right)$ days.

(4) Thermal Stress :



- Due increase in temp., the rod tries to expand against the wall, but is prevented and the compressive stress hence developed is called thermal stress.
- Similarly, due to decrease in temperature, tensile thermal stress develop which prevent the rod to contract.

Reminders : Stiffness corresponding to a rigid body K = $\left(\frac{YA}{l}\right)$

$\therefore$ $\boxed{F = \left(\frac{YA}{l}\right)\Delta l}$ Hence, you can treat it as spring with K.

Expansion in length due to heating = $(l\alpha\Delta\theta)$

Due to stress developed contraction produced $\Delta l = \left(\frac{Fl}{YA}\right)$

But the net expansion is zero. Hence,

$\therefore$ $(\alpha\ l\ \Delta\ \theta) + \left(-\frac{Fl}{YA}\right) = 0$

$\frac{Fl}{YA} = \alpha l \Delta\theta$

$\therefore$ $F = AY\alpha\Delta\theta$

$\therefore$ $\boxed{\text{Thermal Stress} = \left(\frac{F}{A}\right) = \alpha\ Y\Delta\theta}$

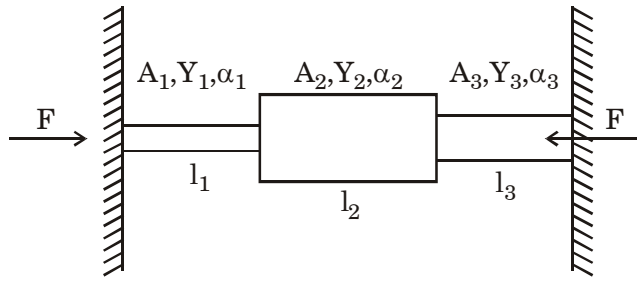
Example 14

Three rods are joined together and kept fixed between two rigid supports. If the temp. of the system is increased by $\Delta\theta$ then find,

(i) force developed at junctions

(ii) length of each rod finally

(iii) displacement of joints.



Solution :

Everywhere, throughout at all functions, same force is developed.

$$\boxed{\Delta L = 0}$$

$$\Rightarrow \quad \Delta l_1 + \Delta l_2 + \Delta l_3 = 0$$

$$\Rightarrow \quad \left(l_1\alpha_1\Delta\theta - \frac{Fl_1}{A_1Y_1}\right) + \left(\alpha_2 l_2\Delta\theta - \frac{Fl_2}{A_2Y_2}\right) + \left(\alpha_3 l_3\Delta\theta - \frac{Fl_3}{A_3Y_3}\right) = 0$$

$$\Rightarrow \quad (l_1\alpha_1 + l_2\alpha_2 + l_3\alpha_3)\Delta\theta = F\left(\frac{l_1}{A_1Y_1} + \frac{l_2}{A_2Y_2} + \frac{l_3}{A_3Y_3}\right)$$

(i) $$\boxed{F = \frac{(\alpha_1 l_1 + \alpha_2 l_2 + \alpha_3 l_3)\Delta\theta}{\left(\frac{l_1}{AY_1} + \frac{l_2}{AY_2} + \frac{l_3}{A_3Y_3}\right)}}$$

(ii) To find the length of each rod finally, use,

$$(l_1)_f = \left(l_1\alpha_1\Delta\theta - \frac{Fl_1}{A_1Y_1}\right) \text{ [F obtained from (i)]}$$

and hence similarly for l_2, l_3

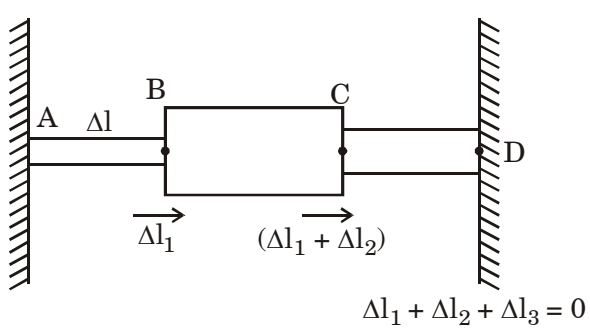
(iii) displacement of each junction

$(\Delta x)A = 0$

$(\Delta x)B = \Delta l_1$

$(\Delta x)C = \Delta l_1 + \Delta l_2$

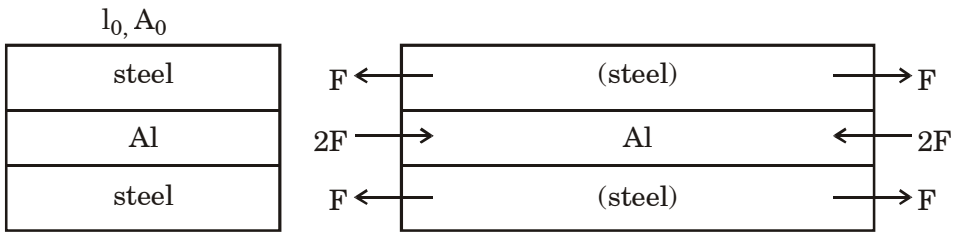
$(\Delta x)D = \Delta l_1 + \Delta l_2 + \Delta l_3 = 0$



Example 15

Two steel rods and one aluminium rod of equal length and cross-sectional area, l_0 and A_0, are riveted together at their ends.

Assume all the rods are in the state of zero tension at 0 C. Find the length of each rod after the temp. is increased by $\Delta\theta$.



Solution :

Now, equate final length of each of the rods.

$$l_{Al} = l_{steel}$$

$$\Rightarrow \quad l_0 + \Delta l_{Al} = l_0 + \Delta l_{steel}$$

$$\therefore \quad (\Delta l_{Al} = \Delta l_{steel})$$

$$\Rightarrow \quad (\alpha_{Al}\, l_0\, \Delta\theta) - \left(\frac{2F\, l_0}{AY_{Al}}\right) = (\alpha_{steel} l_0\, \Delta\theta) + \left(\frac{Fl_0}{AY_{steel}}\right)$$

$$\Rightarrow \quad (\alpha_{Al}\, \Delta\theta) - \left(\frac{2F}{AY_{Al}}\right) = (\alpha_{steel}\, \Delta\theta) + \frac{F}{AY_{steel}}$$

$$\Rightarrow \quad \frac{A(\alpha_{Al}\ \alpha_{steel})\, \Delta\theta}{\left(\frac{2}{Y_{Al}} + \frac{1}{Y_{steel}}\right)} = F$$

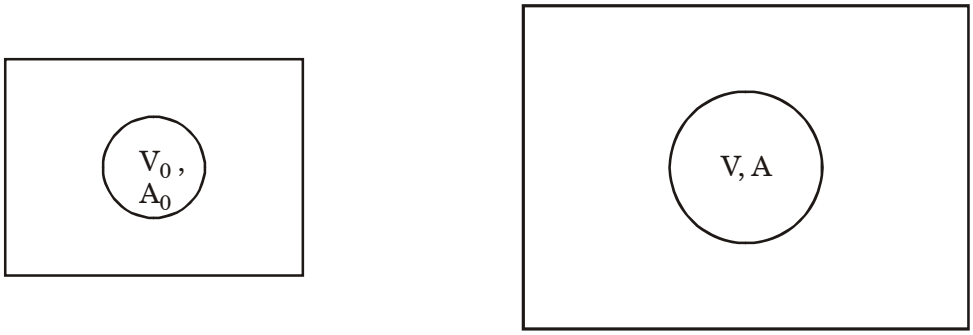
Now put the value of F in

$$\Delta l_{Al} = \alpha_{Al}\, l_0 \Delta\theta - \frac{2F\, l_0}{AY_{Al}}$$

and $\Delta l_{steel} = \alpha_{steel}\, l_0\, \Delta\theta + \dfrac{F\, l_0}{AY_{steel}}$

(5) Thermal expansion of an isotropic body is imagined as a photographic enlargement.

i.e., if there is a hollow cavity (in case of solids) or hole (in case of lamina), then volume of the cavity (or hole) will increase, just like if it were supposed to be filled with the same material.



$$V = V_0 (1 + \gamma\Delta\theta) \rightarrow \text{in case of cavity}$$
$$A = A_0 (1 + \beta\Delta\theta) \rightarrow \text{in case of hole}$$

(6) Relative charge in the volume of the liquid :

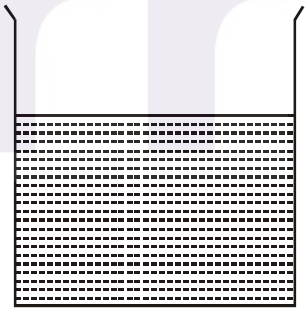
$V_L = V_0 (1 + \gamma_L \Delta\theta)$

$V_S = V_0 (1 + \gamma_S \Delta\theta)$

Change in volume, $\Delta V = V_L - V_S$

$= (\gamma_L - \gamma_S) V_0 \Delta\theta$

$\Delta V = (\gamma \text{ apparent}) V_0 \Delta\theta$



$$\gamma \text{ apparent} = \gamma_L - \gamma_S$$

(1) If $\gamma_L > \gamma_s$

$\Rightarrow$ $\Delta V > 0$ (liquid level will rise)

(2) If $\gamma_L < \gamma_s$

$\Rightarrow$ $\Delta V < 0$ (liquid level will fall)

(3) If $\gamma_L = \gamma_s$

$\Delta V = 0$ [No change in liquid level]

(7) Effect on density

$$\rho_0 = \frac{m}{V_0}, \rho = \frac{m}{V} = \frac{m}{V_0(1+\gamma\theta)}$$

$$\therefore \quad \frac{\rho}{\rho_0} = \frac{1}{1+\gamma\theta}$$

$$\therefore \quad \boxed{\rho = \frac{\rho_0}{1+\gamma\theta}}$$

Further,

$\rho = \rho_0 (1 + \gamma\theta)^{-1} = \rho_0 (1 - \gamma\theta)$

$\rho = \rho_0 (1 - \gamma\theta)$ [This is an approx result]

Note : Also, sometimes in some questions, you do not get correct answer by just applying the above eqn. for a given temp. range.

In that case, start for initial condition at T = 0 C

eg. Correlate ρ_{20} and ρ_{70}

$\rho_{70} = \rho_{20} (1 - \gamma 50) \rightarrow$ This may not yield correct

In that case use,

$\rho_{20} = \rho_0' (1 + \gamma 20)$

$\rho_{70} = \rho_0 (1 + \gamma\ 70)$

$$\therefore \quad \frac{\rho_{70}}{\rho_{20}} = \frac{(1+70\gamma)}{(1+20\gamma)}$$

(8) Effect on Apparent Weight :

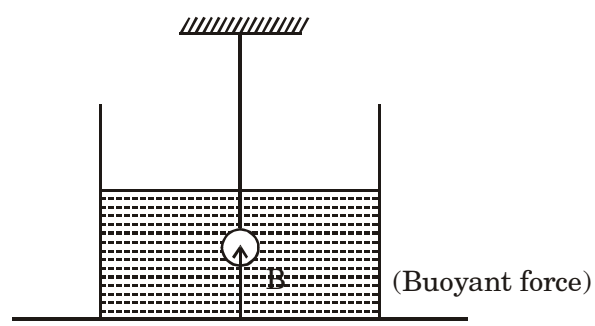
$\boxed{W' = W - B}$ Apparent Wt.

$B_0 = \rho_0 V_0 g$

$B = \rho V g$

$$\therefore \quad \frac{B}{B_0} = \frac{\rho V}{\rho_0 V_0} = \frac{\rho_0(1-\gamma_L\Delta\theta)}{\rho_0} \times \frac{V_0\ (1+\gamma_s\ \Delta\theta)}{V_0}$$

$$\therefore \quad \frac{B}{B_0} = (1-\gamma_L\Delta\theta)(1+\gamma_s\Delta\theta)$$



$$\frac{B}{B_0} = \left[1 - (\gamma_L - \gamma_s)\Delta\theta - \gamma_L\ \gamma_s\ \Delta\theta^2\right]$$

$$\boxed{B = B_0\ [1 - (\gamma_2 - \gamma_s)\Delta\theta}$$ $\{\gamma_L\ \gamma_s\ (\Delta\theta)^2$ can be neglected$\}$

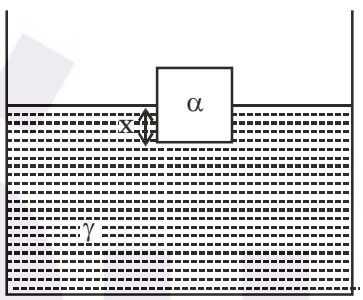
$\gamma_L > \gamma_s$ generally

$\therefore$ $B < B_0$

$\therefore$ with the increase in temp., buoyant force decreases and hence the apparent weight increases.

Example 16

A cubical block is floating in a bath with length x, immersed in liquid. When temp. is increased by $\Delta\theta$, the block keeps floating with same length immersed in liquid. Find the relations between α and γ.



α = coefficient of Linear exxpansion of cube

γ = coefficient of volume of Liquid

Solution :

B_0 = Initial buoyant force

$= (\rho_0\ A_0\ x\ g) = mg$

$B = (\rho\ A\ x\ g) = mg$

$\therefore$ $B = B_0$

$\Rightarrow$ $\rho\ A\ xg = \rho_0 A_0\ x\ g$

$\Rightarrow$ $\rho_0\ (1 - \gamma\Delta\theta)\ A_0\ (1 + 2\alpha\ \Delta\theta) = \rho_0 A_0$

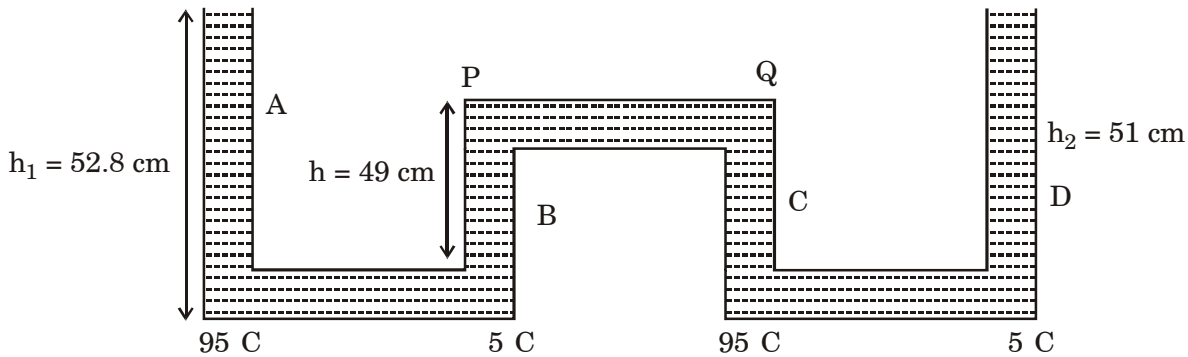
$1 + (2\alpha - \gamma)\ \Delta\theta = 1$

$$\boxed{2\alpha = \gamma}$$

$\gamma = 2\alpha$ **Ans.**

Example 17

In the apparatus shown, find the γ of the liquid.



Solution :

Equate pressure in a horizontal level in stationary liquid.

$$P_P = P_Q$$

$$\Rightarrow \quad P_{atm} + h_1\rho_{95}\ g - h\rho_5\ g = P_{atm} + h_2\ \rho_5\ g - h\rho_{95}\, g$$

$$\rho_{95}\ [h_1 + h] = \rho_5\ (h + h_2)$$

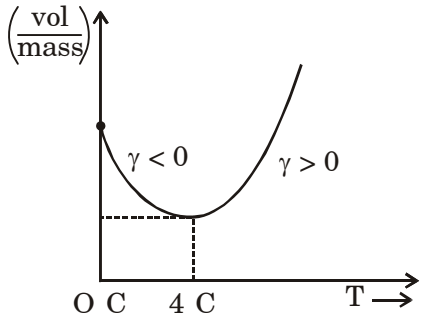
$$\Rightarrow \quad \frac{\rho_{5^\circ}}{(1 + 90\gamma)} = \rho_{5^\circ}\ \frac{(h + h_2)}{(h + h_1)}$$

$$\therefore \quad (1 - 90\gamma) = \frac{51 + 49}{52.8 + 49} = \frac{100}{101.8}$$

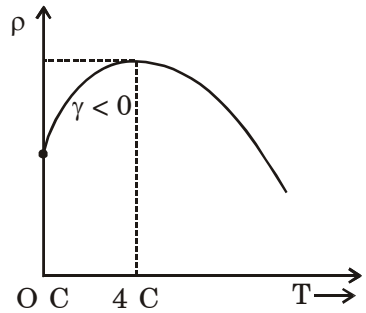
$$\therefore \quad 90\gamma = \left(1 - \frac{100}{101.8}\right) = \frac{1.8}{101.8}$$

$$\therefore \quad \gamma = \frac{1.8}{90 \times 101.8}$$

Anamolous behaviour of water



Minimum volume at 4 C



Maximum density at 4 C

Calorimetry

(1) Heat Capacity (Thermal capacity) C′ :

The amount of heat required to raise the temperature of a substance of 1 C (or 1 K)

$$C' = \frac{\Delta Q}{\Delta T} \quad \Rightarrow \quad \Delta Q = C'\Delta T \text{ (Avg. value)}$$

But, strictly C′ is not constant for a temperature change.

Unit : J/ C or J/K

Absolute Value : $\left(C' = \frac{dQ}{dT}\right)$

$\therefore \quad dQ = C'dT$

$\therefore \quad \boxed{\Delta Q = \int_{T_1}^{T_2} C'dT}$ Take the limits in given range with proper scale (0 C or K)

(2) Molar Heat Capacity (C) :

The amount of heat required to increase the temp. of 1 mol of a substance of 1 C or 1 K.

$$\boxed{C = \frac{\Delta Q}{n\Delta T}} \quad \Rightarrow \quad \Delta Q = nC\,\Delta T(\text{Avg. value})$$

Absolute value $\Delta Q = \int_{T_1}^{T_2} nC\,dT$ units : J/mol/K

(3) Specific Heat Capacity (S) :

The amount of heat required to raise the temperature of unit mass of a substance by 1 C (or 1 K).

$$\boxed{S = \frac{\Delta Q}{m\Delta T}} \Rightarrow \Delta Q = ms\Delta T \quad \text{(Avg. value)}$$

$$\boxed{\Delta Q = \int_{T_1}^{T_2} msdT} \quad \text{(Absolute value) unit : J/kg/K}$$

Relation between C, C′ and S :

C′ = mS = nC

(i) Thermal capacity = (mass) (specific heat capacity)

= (moles) (molar heat capacity)

(ii) C = MS

Molar heat capacity = (molecular mass) (specific heat capacity)

(4) Water Equivalent (W) :

It is the mass of water to which if same amount of heat is given, cause the same temp. change as that of the body.

C′ = m S

= (mass) (specific heat of water)

(Replace the body by same amount of water)

(5) Latent Heat : (Heat of Transformation) (L) :

The amount of heat required to change the state of unit mass of a substance at constant temperature is called latent heat.

$$L = \frac{\Delta Q}{m} \qquad \text{unit : (J/Kg)}$$

$$\therefore \boxed{\Delta Q = mL}$$

(1) **Latent Heat of Fusion (L_F) :**

The amount of heat required to change the state of unit mass of substance from solid to liquid form at its melting point is called latent heat of fusion.

(2) **Latent Heat of Vaporisation :**

The amount of heat required to change the state of a unit mass of substance from liquid to vapour form at its boiling point is called latent heat of vaporisation.

Principle of Calorimetry :

Total heat lost by system = Total heat gained by system [for an isolated system]

Example 18

An iron cube is initially heated to a high temperature and placed on a large ice cube at 0 C. The block melts the ice below it, displaces water and sinks. In the final equilibrium, the upper surface of iron cube just goes inside the ice. Then find the initial temperature of the block.

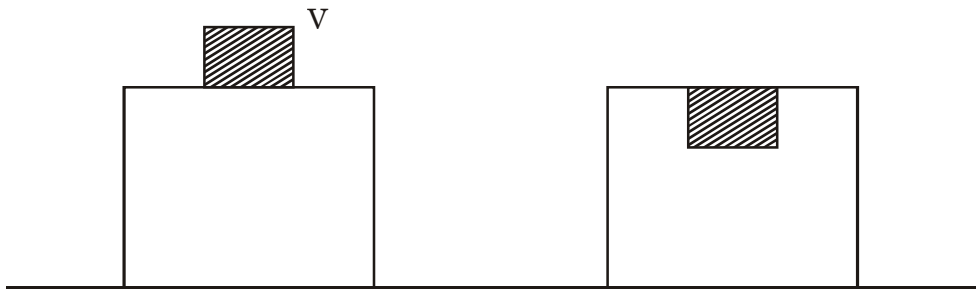
ρ_{iron} **= 8000 kg/m^3**

S_{iron} = 470 J/kg/K

ρ_{ice} = 900 Kg/m^3

L_F = 3.36 10^5 J/kg.

Solution :



Let the initial temperature of the iron cube be θ.

$$\text{Heat lost by cube} = m\theta$$
$$= (\rho_{iron})\ V\ S_{iron}$$
$$\text{Heat gained by ice} = mL$$
$$= (\rho_{ice})\ V\ L_{ice}$$
$$\therefore\ \rho_{iron}\ V\ S_{iron}\ \theta = \rho_{ice}\ V\ L_{ice}$$
$$\therefore\ \theta = \frac{\rho_{ice} \times L_{ice}}{\rho_{iron} \times S_{iron}} = \frac{900 \times 3.36 \times 10^5}{8000 \times 470}$$
$$\theta = \simeq 80\ C$$

Example 19

1 kg of ice at 0 C is mixed with 1 kg of steam at 100 C. Then find the composition of mixture in equilibrium

L_F = 3.36 10^5 J/kg

L_V = 2.26 10^6 J/kg

S_W = 4200 J/kg/K

Solution :

In such question, we have to consider three cases :

Case (i) : The final temp. is 0 C

The entire steam condenses to water at 0 C.

Case (ii) : The entire ice converts to water at 100 C.

Finally temp. : 100 C.

Case (iii) : Final temp is θ where $0 < \theta < 100$ C

In this case, it can be easily decided that final temp. of the mixture is 100 C.

Now, heat gained by ice is converting from ice to water at 0 C and then water at 100 C.

$= mL + mS\Delta\theta$

$= 1 \quad 3.36 \quad 10^5 + 1 \quad 4200 \quad 100$

$= 7.50 \quad 10^5$ J

Total heat that can be lost by steam $= mL_v = 1 \quad 2.26 \quad 10^6$ J

$\therefore$ Total heat that can be lost by steam to convert to water at 0 C > heat required

mass of steam condensed $= \left(\dfrac{7.56 \times 10^5}{2.26 \times 10^6}\right) = 0.335$ kg

$\therefore$ the final comp. of mixture = 1.335 kg of water at 100 C + 0.665 kg of steam

---

Example 20

An ice cube of mass 0.1 kg at 0 C is placed in a thermally insulated container at 227 C. Given that the specific heat of container varies as S = A + BT, where A, B are constants, then find the mass of the container if final temperature is 27 C.

$L_f = 8 \quad 10^4$ cal/kg, S = 1000 cal/kg/K,

A = 100 cal/kg/K

B = 2 $\quad 10^{-2}$ cal/kg/K^2

Solution :

Heat gained by ice in converting to water at 0 C and then to water at 27 C.

$\Delta Q_1 = mL_f + mS\Delta\theta$

$= 0.1 \quad 8 \quad 10^4 + 0.1 \quad 1000 \quad 27$

Heat lost by container

$$= -\int_{T_1}^{T_2} mS\,dt$$

$$= -\int_{500}^{300} m(A + BT)\,dT = m\int_{500}^{300} (A + BT)\,dT$$

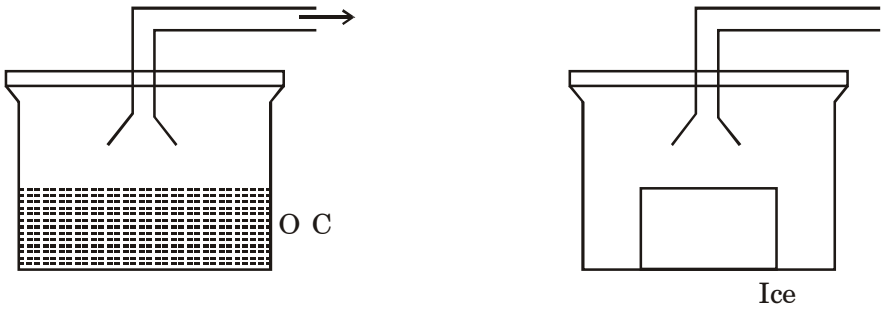
$$\therefore \quad m\int_{500}^{300} (A + BT)\,dT = \left(0.1 \times 8 \times 10^4 + 0.1 \times 1000 \times 27\right)$$

m = 0.495 kg.

Example 21

A thermally isolated container contains 100 gm of water at 0 C. When the air above the water surface is pumped out, some of the water evaporates and remaining freezes. Then find percentage of vapour evaporated.

Solution :



Concept : Water evaporates due to decreases in pressure above the water surface.

Since B.P. is the temperature at which vapor pressure = Patm

$\therefore$ due to decrease in VP, water evaporates

Heat gained by water in evaporation

= Heat lost by water which converts it to ice

$\Rightarrow$ Let x be the amount of water evaporated

$\Rightarrow$ $x \quad L_V = (100 - x) \quad L_f$

$\Rightarrow$ $540\ x = 80\ (100 - x)$ [x is cal]

$\Rightarrow$ $620x = 80 \quad 100$

$$x = \frac{80 \times 100}{620}$$

$$x = \left(\frac{400}{31}\right)$$

$\therefore$ $x \simeq 13$ gm

$\therefore$ Fraction = $\frac{x}{100} \simeq 0.13$

$$mL_v = (M - m)L_f$$

$$\frac{m}{M} = \frac{L_f}{L_v + L_f}$$

$\therefore$ $\frac{\text{evaporated}}{\text{condensed}} = \left(\frac{L_f}{L_v + L_f}\right)$, $\frac{\text{condensed}}{\text{evaporated}} = \left(\frac{L_v}{L_f + L_v}\right)$

Example 22

A loaded glass bulb weighs 156.25 g in air. When the bulb is immersed in a liquid at temperature 15 C, it weighs 57.5 g. On heating the liquid, the apparent weight of the bulb becomes 58.57 g. Find the coefficient of real expansion of the liquid. (Given, coefficient of linear expansion of glass = 9 10^{-6}/ C).

Solution :

When the bulb is immersed in liquid, the loss in its weight is due to upthrust of liquid and is equal to weight of liquid displaced.

Let m_1 and m_2 represents the loss in weight of bulb at lower and higher temperatures t_1 and t_2 respectively; v_1 and v_2 the volumes of bulb and ρ_1, ρ_2 the densities of the liquid at these temperatures respectively. Then,

$$m_1 = \rho_1 v_1 \text{ and } m_2 = \rho_2 v_2$$

$$\therefore \quad \frac{\rho_1}{\rho_2} = \frac{m_1}{m_2} \cdot \frac{v_2}{v_1} \qquad ...(1)$$

If γ_g, is coefficient of volume expansion of bulb,

$$v_2 = v_1[1 + \gamma_g(t_2 - t_1) \qquad ...(2)$$

and

$$\frac{\rho_1}{\rho_2} = [1 + \gamma_R(t_2 - t_1)] \qquad ...(3)$$

Substituting for $\frac{v_2}{v_1}$ and $\frac{\rho_1}{\rho_2}$ from eqns. (2) and (3) in (1),

$$[1 + \gamma_R(t_2 - t_1) = \frac{m_1}{m_2}\left[1 + \gamma_g(t_2 - t_1)\right] \qquad ...(4)$$

Here

$$\gamma_g = 3\alpha_g = 27 \quad 10^{-6}/\text{ C}$$
$$m_1 = 156.25 - 57.5 = 98.75 \text{ g}$$
$$m_2 = 156.25 - 58.57 = 97.68 \text{ g}$$
$$t_1 = 15 \text{ C}, \; t_2 = 52 \text{ C}$$

Substituting in (4),

$$[1 + \gamma_R(52 - 15) = \frac{98.75}{97.68}[1 + 27 \quad 10^{-6} \quad (52 - 15)]$$

Solving $\gamma_R = 32.4 \quad 10^{-5}/\text{ C}$ **Ans.**

KINETIC EQUATION AND EQUIPARTITION OF ENERGY

Consider Na molecules, each of mass m enclosed in a spherical volume as shown.

Now the molecules moving with a velocity v as shown collide with the walls of container. Hence their change in momentum is given by

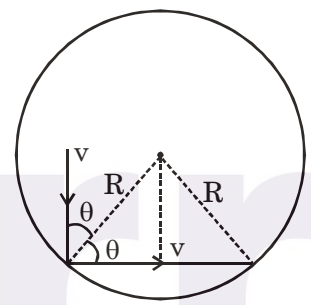
$$\Delta p = (2mv \cos \theta)$$

the time between 2 collisions with the wall of the container can be given by $\Delta t = \left(\frac{2R\cos\theta}{v}\right)$

$$\therefore \quad F_1 = \left(\frac{\Delta p}{\Delta t}\right) = \frac{2mv\cos\theta}{2R\cos\theta}v = \left(\frac{mv^2}{R}\right)$$

$$\text{Total force exerted} = \frac{m}{R}\left(v_1^2 + v_2^2 + v_3^2 + \ldots v_n^2\right)$$

$$= \frac{m \times N}{R}\left(v_{rms}^2\right)$$



$$\therefore \quad \text{Pressure exerted} = \frac{m \times N}{\left(4\pi R^2\right).R}v_{rms}^2$$

$$P = \frac{(m\,N)v_{rms}^2}{\left(4\pi R^3\right)}$$

$$P = \frac{(mN)}{3 \times \left(\frac{4}{3}\pi R^3\right)}v_{rms}^2 = \frac{1}{2}\left(\frac{mN}{V}\right)v_{rms}^2 \quad \text{(Kinetic equation of gas)}$$

$$\boxed{P = \frac{1}{3}\left(\frac{mN}{V}\right)v_{rms}^2 = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3}\left(\frac{M'}{V}\right)v_{rms}^2}$$

m = mass of each molecule M´ = mass of gas

N = no. of molecules V = volume of gas

$$\therefore \quad v_{rms} = \sqrt{\left(\frac{3P}{\rho}\right)} = \sqrt{\frac{3RT}{M(\text{in kg})}} = \sqrt{\frac{3\,KT}{m}}$$

where $K = \left(\frac{R}{N_A}\right) \equiv$ Boltzmann constant = (1.38 10^{-23} J/K)

$$v_{avg} = \left(\frac{v_1 + v_2 + \dots v_n}{N}\right) = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8P}{\pi\rho}} = \sqrt{\frac{8KT}{\pi m}}$$

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{\rho}} = \sqrt{\left(\frac{2KT}{m}\right)}$$

$$\boxed{v_{rms} > v_{avg} > v_{mp}}$$

Avg. translational KE of each molecule

$$= \frac{\frac{1}{2} m\left(v_1^2 + v_2^2 + \dots v_n^2\right)}{N}$$

$$= \frac{1}{2} m\, v_{rms}^2 = \frac{1}{2} m\left(\frac{3KT}{m}\right) = \left(\frac{3}{2}KT\right)$$

$\therefore$ Avg. translational KE of each molecule = $\left(\frac{3}{2}KT\right)$

Now, $\frac{3}{2}KT = \frac{1}{2} m\left(v_{rms}^2\right)$

$$= \frac{1}{2} m \frac{[(v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots v_x^2) + \left(v_{1y}^2 + \dots + v_{ny}^2\right)]}{N}$$

Now, due to random motion

$$\left[< v_x^2 > = < v_y^2 > = < v_z^2 >\right]$$

$$\Rightarrow \quad \frac{3}{2}KT = \frac{1}{2} m < v_x^0 > + \frac{1}{2} m < v_y^2 > + \frac{1}{2} m < v_z^2 >$$

$$\therefore \quad \frac{1}{2} m < v_x^2 > = \frac{1}{2} m < v_y^2 > \frac{1}{2} < v_z^2 > = \left(\frac{1}{2}KT\right)$$

Law of Equipartition of Energy

On average, $\left(\frac{1}{2}KT\right)$ energy can be associated with each degree of freedom.

Degree of Freedom :

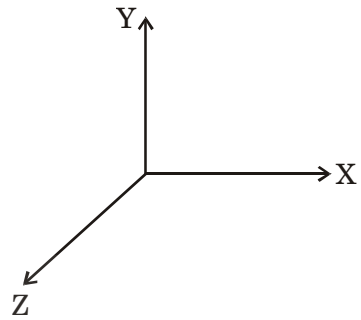
The number of independent modes of motion of a physical system is called degree of freedom.

(1) For monoatomic gas :

f = total degree of freedom

f = 3 translational

$\therefore$ $\boxed{f = 3}$



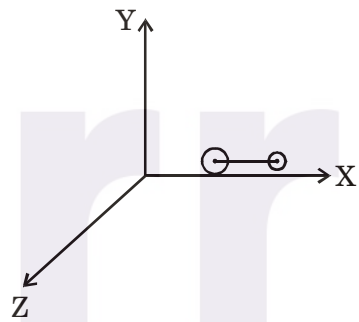
(2) For diatomic gas (also for linear polyatomic)

f = 3 trans + 2 rotational

(rotation along the bond axis) is neglected

$\therefore$ $\boxed{f = 5}$ $\rightarrow$ at ordinary temp.

But at high temp., f = 7



Reason : 2 degree of freedom are also contributed by vibrational motion.

For high temperature, vibrational motion cannot be neglected, hence it must also be counted.

At high temp.

f = 3 trans. + 2 rotat + 2 vibrat. = 7

(3) For polyatomic gas (Non-Linear)

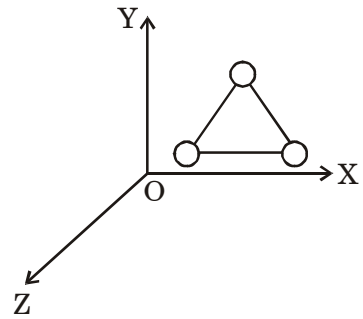
f = 3 trans + 3 rotations

= 6

$\boxed{f = 6}$ At low temp.

At high temp. f = 3 trans + 3 rotat + 2 vib.

$\boxed{f = 8}$



Relation between C_P, C_V and γ

$$U = f\left(\frac{1}{2}KT\right) \rightarrow \text{for 1 molecule}$$

$$\text{for n moles of a gas } U = (n\,N_A)\,f\left(\frac{1}{2}KT\right) = n\left(\frac{fR}{2}\,T\right)$$

$$C_V = \frac{dU}{ndT} = \frac{nfR/2}{n\,dT}\,dT = \left(\frac{fR}{2}\right)$$

$$\boxed{C_V = \left(\frac{fR}{2}\right)}$$

$$\boxed{C_P = \left(\frac{fR}{2} + R\right) = \left(\frac{f}{2} + 1\right)R}$$

$$\gamma = \frac{C_P}{C_V} = \frac{(f+2)\,R(2)}{2\,fR} = \left(1 + \frac{2}{f}\right)$$

$$\therefore \quad \gamma = 1 + \frac{2}{f}$$

for monoatomic gas

$$C_V = \frac{3R}{2},\ C_P = \frac{5R}{2},\ \gamma = \left(\frac{5}{3}\right)$$

for diatomic gas (also CO_2) :

$$C_V = \frac{5R}{2}, C_P = \frac{7R}{2}, (\gamma = 7/5)$$

C_P, C_V and γ for a mixture of gases

Let n_1 mol of monoatomic, n_2 mol of diatomic, n_3 mol of gas mixture be present whose temperature is increased by dT at constant volume.

$$(n_1\,C_{V_1}dT + n_2\,C_{V_2}dT + n_3\,C_{V_3}dT) = (n_1 + n_2 + n_3)\,C_{Vmix}$$

$$\therefore \quad \boxed{(C_V)_{mix} = \left(\frac{n_1\,C_{V_1} + n2\,C_{V_2} + n_3\,C_{V_3}}{n_1 + n_2 + n_3}\right)}$$

Similarly, if you increase temperature by dT at constant pressure you get

$$\boxed{(C_P)_{mix} = \left(\frac{n_1 C_{P_1} + n_2 C_{P_2} + n_3 C_{P_3}}{n_1 + n_2 + n_3}\right)}$$

for $(\gamma_{mixture})$,

$$(\gamma_{mixture}) = \frac{C_P\,(mix)}{C_V\,(mix)}$$

or, $$\frac{n_1 R}{(\gamma_1 - 1)} dT + \frac{n_2 (R)}{(\gamma_2 - 1)} dT + \frac{n_3 R dT}{(\gamma_3 - 1)} = (n_1 + n_2 + n_3) \frac{R\,dT}{(\gamma_{mix} - 1)}$$

$$\therefore \quad \boxed{\frac{n_1 + n_2 + n_3}{(\gamma_{mix} - 1)} = \frac{n_1}{(\gamma_1 - 1)} + \frac{n_2}{(\gamma_2 - 1)} + \frac{n_3}{(\gamma_3 - 1)}}$$

Example 23

A mixture contains 28 gm of N_2 and 32 gm of O_2 gas. It is observed that 50% of O_2 gas dissociates into gaseous atoms. Find γ of the mixture.

Solution :

$$O_{2(g)} \rightarrow 2O_{(g)}$$

(0.5 mol) (0.5 mol) 2

= 1 mol

$n_2 = 1 + 0.5 = 1.5$

$$\therefore \quad C_V(mix) = \frac{n_1 C_{V_1} + n_2 C_{V_2}}{(n_1 + n_2)} = \frac{\left(1 \times \frac{3R}{2}\right) + \left(1.5 \times \frac{5R}{2}\right)}{(2.5)}$$

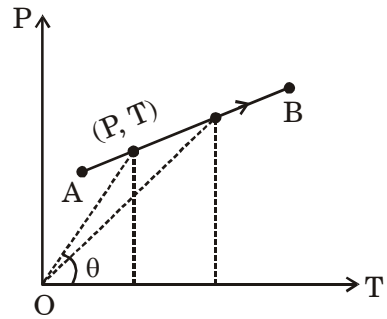
$$= (2.1\ R)$$

$$C_P\,(mix) = C_V\,(mix) + R = 3.1\ R$$

$$\gamma = \left(\frac{3.1}{2.1}\right) \simeq 1.4$$

Important Concepts

(I) From the given graph, decide whether the volume increases or decreases



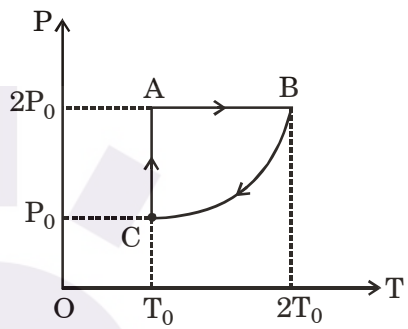
$$\tan\theta = \left(\frac{P}{T}\right) = \left(\frac{nR}{V}\right)$$

$\therefore$ $\left(\text{slope } \alpha \frac{1}{V}\right)$

As slope decreases from A to B,

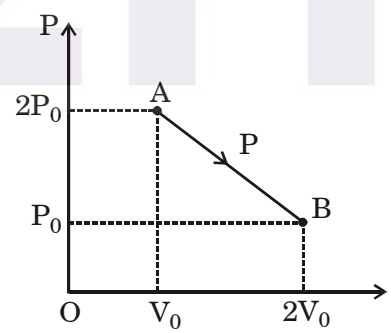
$\therefore$ the volume of the gaseous system undergoing the process increases.

e.g.



Note the from B to C first the volume increase and then decrease (check the slope)

(II) If the PV diagram is a straight line with negative slope then T, V graph is a parabola. Write the equation of the straight line



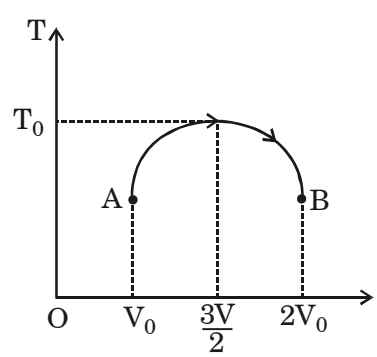
$$P - 2P_0 = \frac{-P_0}{V_0}(V - V_0)$$

$$P = \frac{-P_0 V}{V_0} + 3P_0$$

$$\frac{nRT}{V} = \frac{-P_0 V}{V_0} + 3P_0$$

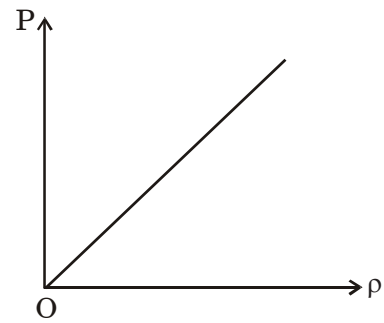
$$\boxed{nRT = \left(\frac{-P_0}{V_0}V^2 + 3P_0 V\right)}$$

$$\therefore \quad \boxed{T = \frac{-P_0 V^2}{nRV_0} + \frac{3P_0 V}{nR}}$$



Illustration

What does this process indicate ?



Solution :

$\rho \ \alpha \ \rho$

$P = K\rho$

$K = \frac{RT}{M}$ = slope of the gas = constant

Hence, T is also constant. Thus it is isothermal process.

(III) Efficiency in a cyclic process :

$$\eta = \left(\frac{\text{work done in cycle}}{\text{net heat absorbed}} \times 100\right)\%$$

$$= \left(\frac{\Delta W_{cycle}}{\Delta Q_{supplied}} \times 100\right)\%$$

$$= \left(\frac{\Delta CW_{cycle}}{\Delta Q_{supplied}} \times 100\right)\%$$

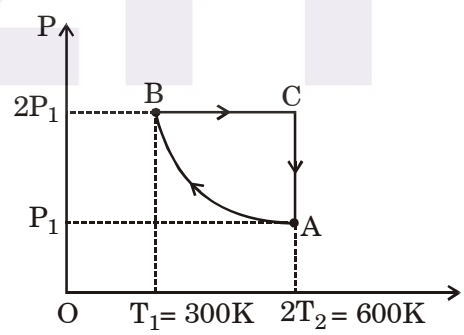
Example 25

2 mol. of an ideal monoatomic gas is taken through a cyclic process, ABC as shown on PT diagram during process AB (PT = const.). Find :

(a) Show the cycle on PV, and VT diagram

(b) Work done by the gas during process AB

(c) Heat absorbed or released during each process. Also find the efficiency of the cycle.



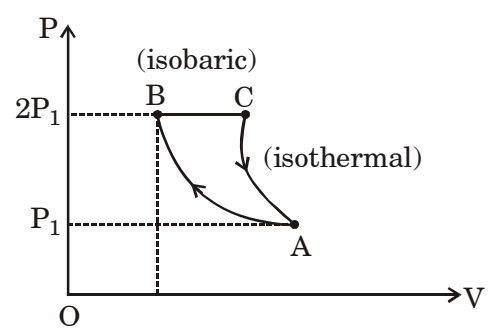
Solution :

For AB : PT = K

$$\frac{P^2V}{nR} = K$$

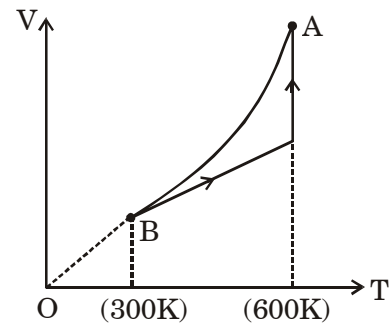
$PV^{1/2}$= K (polytropic process)

$$x = \frac{1}{2}$$



$$\frac{dP}{dV} = \frac{-1}{2} KV^{-3/2} < 0$$

$$\left(\frac{d^2P}{dV^2}\right) > 0 \quad \{\text{To check the slope }\}$$



Now, similarly we can solve for VT curve.

(b) Work done by gas during AB

(PT = K)

$$\therefore \quad W = \int_{V_1}^{V_2} PdV$$

$\therefore$ AB is a polytropic process

$$\therefore \quad \text{Use, } W = \frac{(P_1V_1 - P_2V_2)}{(x-1)} = \frac{(P_2V_2 - P_1V)}{(1-½)}$$

$$\therefore \quad W = 2nR\,(T_2 - T_1) = 2 \quad 2 \quad R \quad 300 \text{ K}$$

$$= 2 \times 2 \times \frac{25}{3} \times 300 = 10^4 \text{ J}$$

(c) Heat evolved or absorbed during process AB,

$$C = \frac{3R}{2} + \frac{R}{(1-½)} = \frac{3R}{2} + 2R = \left(\frac{7R}{2}\right)$$

$$\therefore \quad \Delta Q_{AB} = 2 \times \frac{7R}{2} \times (-300) = -2100 \text{ R}$$

Now $\quad \Delta Q_{BC} = 2 \times \frac{5}{2} \times R \times \Delta T \qquad \left\{R = \frac{25}{3} \text{ J/mol/K}\right\}$

$$= 5 \quad R \quad 300 = 1500 \text{ R}$$

$$\Delta Q_{CA} = nRT \ln \frac{P_1}{P_2} = 840 \text{ R}$$

$$\therefore \quad \text{efficiency } (\eta) = \frac{240 \text{ R}}{2340 \text{ R}} \times 100$$

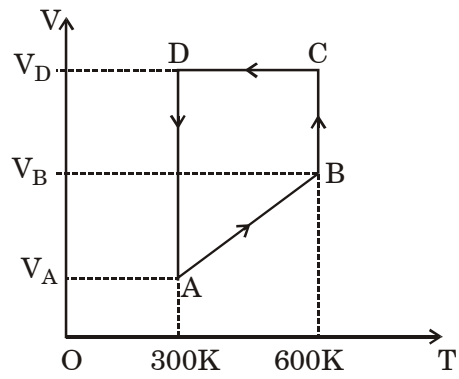
Example 26

2 mol of an ideal monoatomic gas is taken through a cyclic process ABCD as shown on V-T diagram.

(a) Show the cycle on P-V and P-T diagram

(b) Heat evolved (or absorbed) during each process

(c) η of the process



Given that : $\left(\frac{V_D}{V_A}\right) = 4, \left(\frac{V_B}{V_A}\right) = 2$

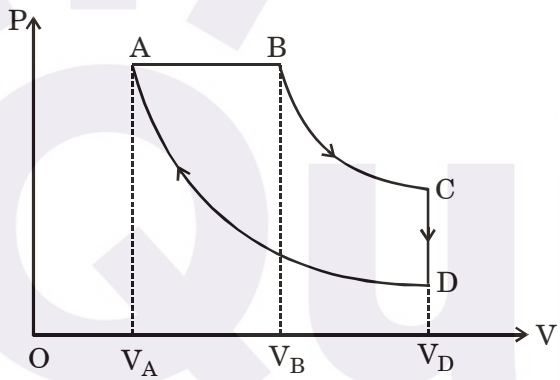
Solution :

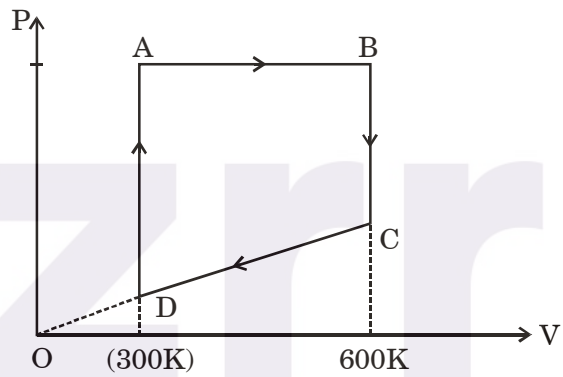
AB is isoboric (V = KT)

BC : isothermal

CD : isochoric

DA : isothermal





(b) $\Delta Q_{AB} = n\left(\frac{5}{2}R\right)(\Delta T) = \frac{2 \times 5 \times R}{2}(300)$

$\therefore \quad \Delta Q_{AB} = 1500\ R\ J$

$\Delta Q_{BC} = nRT\ln\left(\frac{V_D}{V_B}\right) = 2 \quad R \quad 600 \quad \ln 2$

$= 1200R\ln 2\ J$

$\Delta Q_{CD} = nC_V\Delta T = 2 \times \frac{3}{2}R \times (-300) = -900$

$\Delta Q_{DA} = n\ RT\ \ln\left(\frac{V_A}{V_D}\right) = 2 \times R \times 300\ \ln\left(\frac{1}{4}\right)$

$\therefore \quad \Delta Q_{cycle} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CD} + \Delta Q_{DA}$

$= 1500\ R + 1200\ R \ln 2 - 900\ R - 600\ R \ln 4$

$= 1500\ R - 900\ R$

$\therefore \quad \Delta Q_{cycle} = 5000\ J$

(c) $\eta = \dfrac{5000}{1500 \times \dfrac{25}{3} + 1200 \times \dfrac{25}{3} \times 0.7}$

$= \dfrac{5000}{19500}$

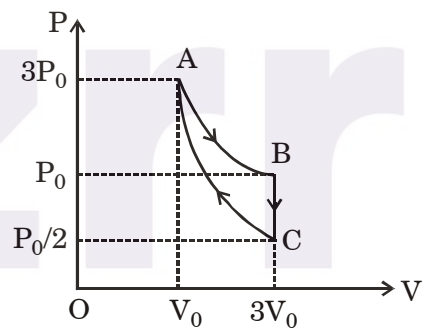
$\eta \simeq 25\ \%$

Tip : When relation between ΔQ, ΔU or ΔW is given then relate ΔQ and ΔU using first law of thermodynamics and use

$\Delta Q = nC\Delta T$

$\Rightarrow \Delta U = nC_V\Delta T$

Example 27



2 mol of an ideal monoatomic gas taken through a cyclic process AB where AB is isothermal and in the process CA, $\Delta Q_{CA} + \Delta W_{CA} = 0$

(a) Then show the cycle on P-T, V-T diagram

(b) Heat absorbed or released during each process

(c) Also find the efficiency

Solution :

For the process CA, use

$\Delta Q_{CA} = -\Delta W_{CA}$

$\Delta Q_{CA} = -(\Delta Q_{CA} - \Delta U_{CA})$ [from first law of thermodynamics]

$2\Delta Q_{CA} = \Delta u_{CA}$

$2nC\Delta T = n\ C_V\ \Delta T$

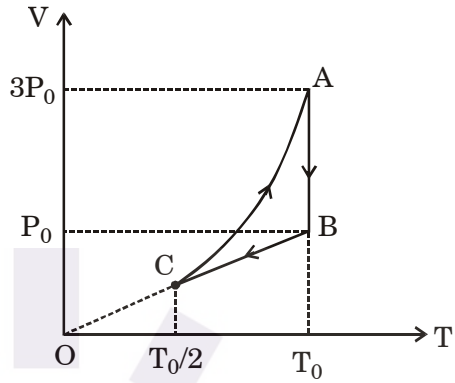
$C = \left(\dfrac{C_V}{2}\right) = \left(\dfrac{3R}{4}\right)$

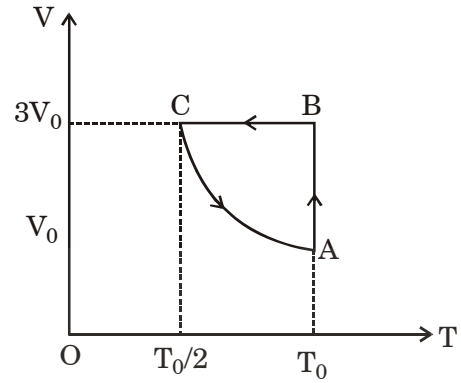
$\therefore \quad C = \left(\dfrac{3R}{4}\right)$

$\therefore$ CA is a polytropic process

$$C = C_V + \frac{R}{(1-x)}$$

$$\frac{3R}{4} = \frac{3R}{2} + \left(\frac{R}{1-x}\right) \quad \Rightarrow \quad x = \left(\frac{4}{3}+1\right) = \left(\frac{7}{5}\right)$$





(b) Heat evolved (or absorbed) during each process

$$\Delta Q_{AB} = nRT \ln\left(\frac{3V_0}{V_0}\right) = 3P_0V_0 \ln 3$$

$$\Delta Q_{BC} = \Delta U = nC_V\Delta T$$

$$= \frac{3}{2}(nR\Delta T) = \frac{3}{2}V\Delta P - \left(\frac{9P_0V_0}{4}\right)$$

$$\therefore \Delta Q_{BC} = -\left(\frac{9P_0V_0}{4}\right)$$

$$\Delta Q_{CA} = nC\Delta T$$

$$= \frac{3}{4}(nR\Delta T) = \frac{3}{4}\left(3P_0V_0 - \frac{3P_0V_0}{2}\right)$$

$$= \frac{3}{4}\left(\frac{3P_0V_0}{2}\right) = \left(\frac{9P_0V_0}{8}\right)$$

(c) η of the cycle

$$= \frac{\left(3P_0V_0\ln 3 - \frac{9V_0P_0}{4} + \frac{9V_0P_0}{8}\right)}{\left(3P_0V_0 \ln 3 + \frac{9V_0P_0}{8}\right)} \times 100 = \left(\frac{3\ln 3 - \frac{9}{8}}{3\ln 3 + \frac{9}{8}}\right) \times 100$$

Example 28

Air initially at 260 C (533 K), P_0 = 700 Pa and V_0 = 0.028 m^3 is expanded isobarically upto $3V_0$ = 0.084 m^3. Then a polytropic process is followed with n = 1.5 and an isothermal process is carried out to complete the cycle.

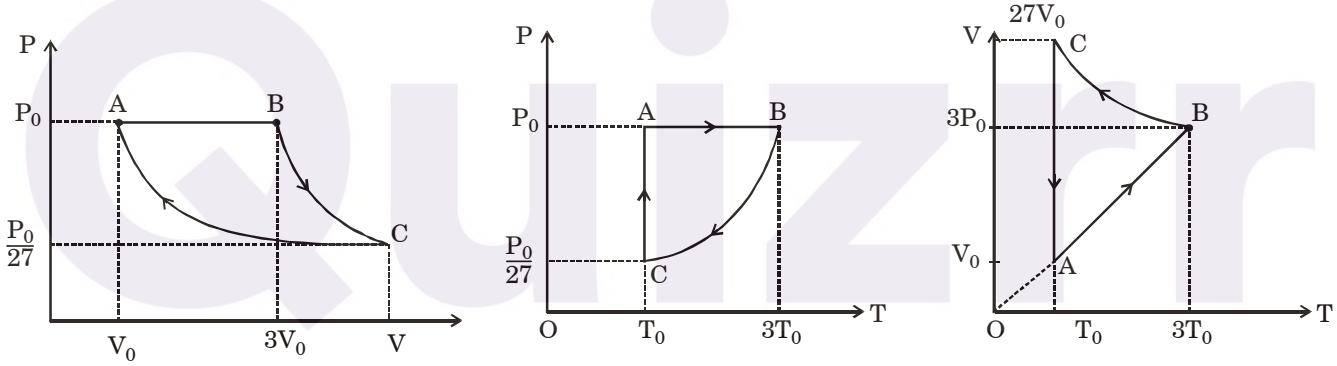
(a) Show the cycle on PV, PT, VT diagram

(b) Heat evolved (or absorbed) during each process

(c) Efficiency of the cycle

Solution :

Remember : Air is diatomic



(b) Heat evolved (or released) during each process

$$\Delta Q_{AB} = n\ C_P \Delta T$$

$$= n\left(\frac{7R}{2}\right)\Delta T = \frac{7}{2}P\Delta V = \frac{7}{2}P_0(2V_0) = 7P_0V_0$$

$$\Delta Q_{BC} = n\ C\Delta T$$

$$\left[C = C_V + \frac{R}{(1-x)} = \frac{5R}{2} + \frac{R}{1 - {}^3/_2} = \frac{5R}{2} - 2R = \left({}^R/_2\right)\right]$$

$$\Delta Q_{BC} = n\left(\frac{R}{2}\right)\Delta T = \frac{P_0V_0 - 3P_0V_0}{2} = (-P_0V_0)$$

$$\Delta Q_{CA} = \Delta W_{CA}$$
$$= nRT \ln (27)^{-1} = nRT \ln (1/27)$$
$$= -P_0V_0 \ln 27$$
$$= (-3P_0V_0 \ln 3)$$

(c) $\eta = \dfrac{6P_0V_0 - 3P_0V_0 \ln 3}{7P_0V_0} \times 100 = \left(\dfrac{6 - 3\ln 3}{7}\right) \times 100$

Example 29

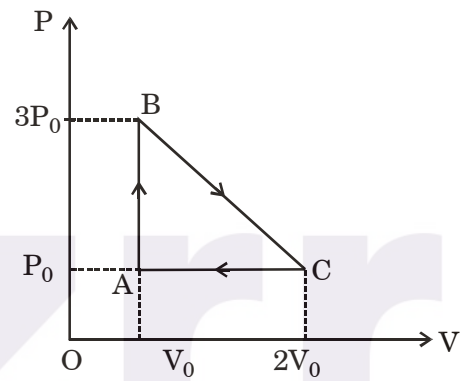
1 mol of an ideal gas is taken through a cycle process as shown in diagram.

(a) Work done by gas per cycle

(b) Heat absorbed (or evolved) during each process

(c) The maximum temperature attained during cycle

Solution :



(a) $\Delta W_{cycle} = \dfrac{1}{2} \times V_0 \times 2P_0 = (P_0V_0)$

$\Delta W_{AB} = 0$

$\Delta W_{CA} = -(P_0V_0)$

$\therefore \quad \Delta W_{BC} = 2P_0V_0$

(b) Heat (absorbed) released during each cycle

$$\Delta Q_{AB} = nC_V\Delta T$$
$$= \frac{3}{2}(nR\Delta T) = \frac{3}{2} \times 2P_0V_0 = (3P_0V_0)$$
$$\Delta Q_{CA} = nC_P\Delta T$$
$$= \frac{5}{2} P_0 (-V_0) = \frac{-5}{2} P_0V_0$$

$\therefore \quad \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} = \Delta W_{cycle}$

$\therefore \quad \Delta Q_{BC} = P_0V_0 - \left(3P_0V_0 - \dfrac{5}{2} P_0V_0\right)$

$$= P_0V_0 - \left(\frac{P_0V_0}{2}\right) = \frac{(P_0V_0)}{2}$$

(c) Note that during the process BC, the temp. initially increases and then decreases

$$(P - 3P_0) = \frac{-2P_0}{V_0}(V - V_0)$$

$$P - 3P_0 = \frac{-2P_0V}{V_0} + 2P_0$$

$$\therefore \quad P = \frac{-2P_0V}{V_0} + 5P_0$$

$$\boxed{\text{Replace } P = \left(\frac{nRT}{V}\right)}$$

$$\therefore \quad \frac{nRT}{V} = \frac{-2P_0V}{V_0} + 5P_0$$

$$\Rightarrow \quad T = \left(\frac{-2V_0P_0}{nRV_0} + \frac{5P_0V}{nR}\right)$$

To find the maximum temperature differentiate the above,

$$\frac{dT}{dV} = \left(\frac{-4P_0V}{nRV_0} + \frac{5P_0}{nR}\right) = 0$$

$$\therefore \quad V = \left(\frac{5V_0}{4}\right)$$

$$\therefore \quad \text{Maximum temp. } T = -\frac{2P_0}{nRV_0}\left(\frac{25V_0^2}{16}\right) + \frac{5P_0}{nR}\left(\frac{5V_0}{4}\right)$$

$$= \frac{P_0V_0}{nR}\left(\frac{-25}{8} + \frac{25}{4}\right) = \frac{25P_0V_0}{8nR}$$

Example 30

An ideal gas is taken through a process in which U varies as U = aV^{α} where a and α are constants. Given, the adiabatic contant of the gas is γ, find

(i) molar heat capacity of the process

(ii) heat required to increase the internal energy of given by ΔV

(iii) Work done during part (ii)

Solution :

(i) $U = aV^{\alpha}$

$nC_VT = aV^{\alpha}$

$$\frac{nRT}{(\gamma-1)} = \alpha V^{\alpha} \qquad \therefore \qquad \frac{PV}{(\gamma-1)} = \alpha V^{\alpha}$$

$\therefore$ $\boxed{P\,V^{1-\alpha} = \alpha\,(\gamma-1) = \text{const}}$

$\therefore$ the process is a polytropic with $x = (1 - \alpha)$

$$\therefore \quad C = \frac{R}{(\gamma-1)} + \frac{R}{\left[1-(1-\alpha)\right]} = \frac{R}{(\gamma-1)} + \left(\frac{R}{\alpha}\right) = R\left[\frac{\alpha+\gamma-1}{\alpha(\gamma-1)}\right]$$

(ii) $\Delta U = n\,C_V\,\Delta T$

$$(\Delta U) = \frac{nR}{(\gamma-1)}\Delta T \qquad \therefore \quad nR\Delta T = (\Delta U)\,(\gamma-1)$$

$\Delta Q = nC\Delta T$

$$= \left[\frac{1}{(\gamma-1)} + \frac{1}{\alpha}\right] nR\Delta T = \left(\frac{1}{\alpha} + \frac{1}{\gamma-1}\right)\Delta U(\gamma-1)$$

$$= \Delta U\left(\frac{\gamma-1}{\alpha} + 1\right)$$

(iii) $\Delta W = \Delta Q - \Delta U$

$$\Delta W = \Delta U\left[\frac{\gamma\text{-}1}{\alpha} + 1\right] - \Delta U = \Delta U\left\{\frac{\gamma-1}{\alpha}\right\}$$

Example 31

1 mol of an ideal gas is taken through a process $T = T_0 + \alpha V$

(i) Find the molar heat capacity of the process

(ii) Heat required to increase the volume of gas from V_1 to V_2

Given that, specific heat of gas at const. process = C_P

Solution :

$T = (T_0 + \alpha V)$

Use $\boxed{C = C_V + \frac{PdV}{ndT}}$

$$\left(\frac{dT}{dV}\right) = \alpha \qquad \therefore \qquad \left(\frac{dV}{dT}\right) = (1/\alpha)$$

$$C = C_V + \left(\frac{P}{n\alpha}\right) = C_V + \frac{nRT}{nV\alpha}$$

$$C = C_V + \frac{nR\left(T_0 + \alpha V\right)}{n\,\alpha V}$$

$$C = C_V + \frac{RT_0}{\alpha V} + R = \left(C_P + \frac{RT_0}{\alpha V}\right)$$

(ii) Heat required to increase the volume from V_1 to V_2

$\therefore \quad \Delta Q = nC\ dT$

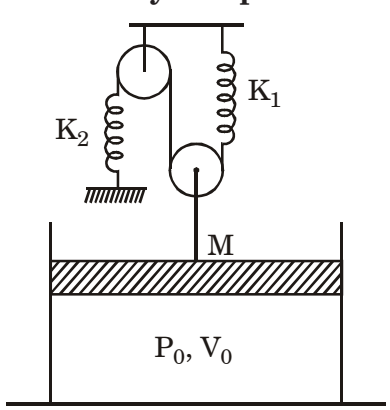
$= nC\ \alpha dV \qquad dT = \alpha dV$

$$= \int_{V_1}^{V_2} \left(C_P + \frac{RT_0}{\alpha V}\right) n\,\alpha\ dV$$

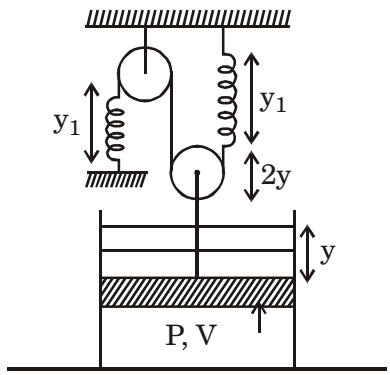
$$\Delta Q = n\alpha\left[C_P\left(V_2 - V_1\right) + \frac{RT_0}{\alpha}\ln\left(\frac{V_2}{V_1}\right)\right]$$

Example 32

An ideal gas with adiabatic const. γ is enclosed in an adiabatic container fitted with an adiabatic piston of mass m which can smoothly move inside the cylinder. The pressure and volume of the gas in equilibrium are P_0 and V_0. If the piston is slightly displaced, then find the time period of the SHM exclude by the piston.



Solution :



$y_1 + y_2 = 2y$

Note : For any polytropic process, we have $\left(\frac{dP}{dV}\right) = -x\left(\frac{P_0}{V}\right)$

Restoring force exerted by the gas :

$$\frac{dP}{dV} = -x\left(\frac{P}{V}\right)$$

$$\therefore \quad dP = -x\left(\frac{P}{V}\right)[-Ady] \qquad \{ dV = -Ady \}$$

$$(dPA) = +\ x\left(\frac{P}{V}\right)A^2 y$$

$\therefore$ force exerted by gas = $\left[x\left(\frac{P_0}{V_0}\right)A^2 y\right]$ (toward equilibrium position)

Now, due to the pulley combination

$y_1 + y_2 = 2y$ ...(1)

Net torque on pulley $P_1 = 0$ [Reason : (I is zero)]

$\Rightarrow \quad (K_1 y_1 = K_2 y_2)$ ...(2)

Increase in tension $\Delta T = (K_1 y_1 + K_2 y_2)$

$= (2\ K_1 y_1)$

from (1) and (2) $\left(y_1 + \frac{K_2 + K_1}{K_2}\right) = 2y$

$$\therefore \quad y_1\left(\frac{K_2 + K_1}{K_2}\right) = 2y$$

$$\therefore \quad DT = 4y\frac{(K_1 K_2)}{(K_1 + K_2)}$$

$\therefore$ Net restoring force

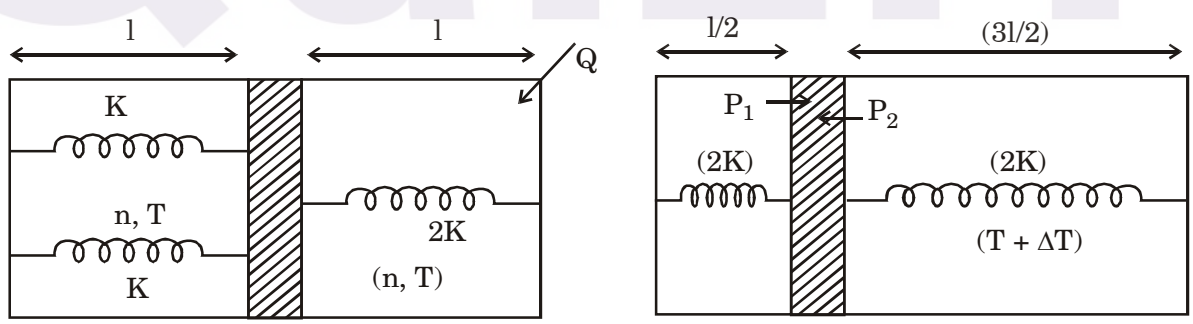
$$= \left(\frac{x\,P_0\,A^2}{V_0}y\right) + \left(\frac{4y\,K_1\,K_2}{K_1 + K_2}\right) = ma = m\left(\frac{d^2y}{dt^2}\right)$$

$$\therefore \quad \left(\frac{d^2y}{dt^2}\right) = -\frac{1}{m}\left(\frac{xP_0A^2}{V_0} + \frac{4K_1K_2}{K_1 + K_2}\right)y$$

$$\therefore \quad T = 2\pi\sqrt{\frac{m}{\left[\frac{xP_0A^2}{V_0} + \left(\frac{4K_1\,K_2}{K_1 + K_2}\right)\right]}}$$

Example 33

A horizontal cylinder of length 2 l in initially divided by a thin insulating piston into 2 equal parts, each of which contain n mol of an ideal gas (monoatomic) at equal temp. T. Now a heat Q is supplied to the right chamber and piston displaces by a distance $\left(\frac{l}{2}\right)$ towards left. Then find the heat dissipated by left chamber gas during the process if temp. of left chamber remains const. and there is no heat exchange through right chamber.



Solution :

Let the heat dissipated through left chamber be Q. Now the left chamber is undergoing isothermal compression

$\therefore \quad Q' = \Delta W$

Concept : Work done by the right chamber gas is used in increasing the potential energy of the spring and on work done on the left chamber

$$Q = \Delta u + \Delta W$$

$$Q = nC_V\Delta T + \left[\left(\frac{1}{2}\frac{kl^2}{4}\right)\times 2 + Q'\right]$$

$$\therefore \quad Q' = \left(Q - nC_V\Delta T - \frac{kl^2}{4}\right)$$

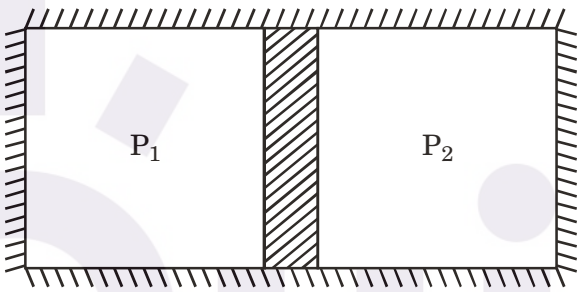
At equilibrium position, $P_2A = P_1A\,(kl) \therefore (P_2 - P_1)\,A = Kl$

Note :

(1) Whenever a piston is moving slowly or is given as light then net force on piston = 0 and

(2) work done by all the forces on piston = 0

(3)



If in an adiabatic compression, the piston moves inside the container, then work done by the gas may or may not be zero.

$dW = (P_2 - P_1)\,Adx$, $dW = -\,dU$

$$\therefore \quad \left[\frac{nR(T+\Delta T)}{V_2} - \frac{nRT}{V_1}\right]A = Kl$$

$$\Rightarrow \quad \left[\frac{nR(T+\Delta T)}{A(3l)/2} - \frac{nRT}{A(l/2)}\right]A = Kl$$

$$\Rightarrow \quad \frac{nR}{l}\left[\frac{2T}{3} - 2T + \frac{2\Delta T}{3}\right] = Kl$$

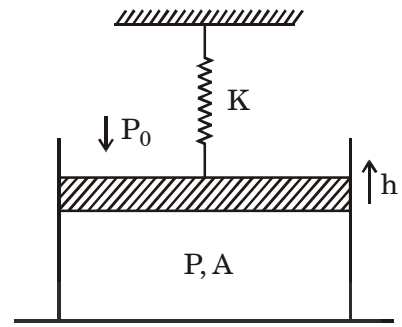
$$\therefore \quad \Delta T = \frac{Kl^2}{nR} + \left(\frac{4T}{3}\right)$$

$$\therefore \quad Q' = Q - n\left(\frac{3}{2}R\right)\left[\frac{Kl^2}{nR} + \left(\frac{4T}{3}\right)\right] - \left(\frac{Kl^2}{4}\right)$$

$$Q' = Q - \frac{7Kl^2}{4} - (2nRT)$$

Example 34

Find the work done by the gas is moving the piston slowly in the given arrangement by height h.



Solution :

Since, the piston is moved slowly, net force piston = 0

Consider a moment when the piston has displaced up by y

$$mgh + ky + P_0A = PA$$

$$\therefore \quad W = \int_0^h PA.\,dy$$

$$= \int_0^h (mg + ky + P_0A)\,dy$$

$$\Delta W = mgh + \frac{1}{2}kh^2 + P_0Ah$$

Alternative method :

Work done by all forces on piston = $\Delta K = 0$

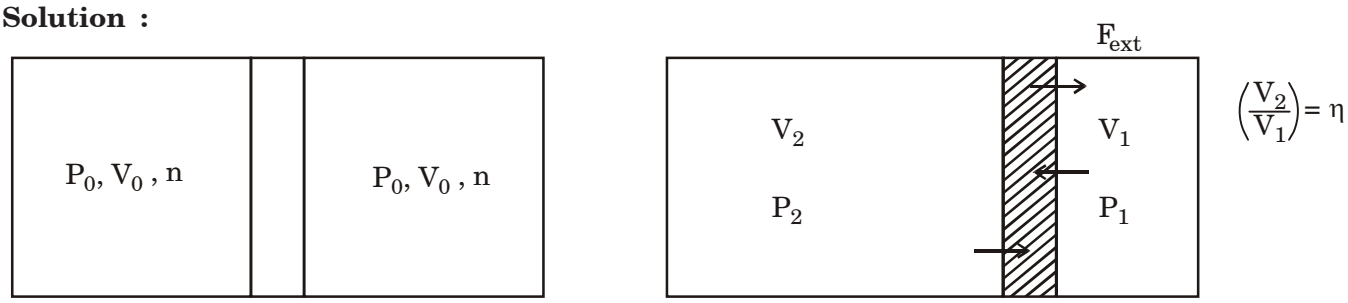
$$-\Delta W + mgh + P_0A\,h + \frac{1}{2}kh^2 = 0$$

$$\therefore \quad \Delta W = mgh + P_0Ah - \frac{1}{2}kh^2$$

Example 35

A thin piston can freely move inside a horizontal container at both ends, initially divided into two equal parts each of them contains ideal gas at equal pressure, volume and moles. Find the work required to increase the volume of larger section by η times than that of the smaller section, by slowly moving the piston isothermally.

Solution :



$$\text{Net force on piston} = 0$$

$$F_{ext} = (P_2 - P_1)A$$

$$\therefore \quad \Delta W_{ext} = \int (P_2 - P_1) A\, dx$$

Now, volume total $V_1 + V_2 = 2V_0$ ...(1)

and $V_1 + \eta V_1 = 2V_0$

$$V_1 (1 + \eta) = 2V_0$$

$$\therefore \quad V_1 = \left(\frac{2V_0}{1+\eta}\right) \quad \therefore \quad V_2 = \left(\frac{2\eta V_0}{1+\eta}\right) \quad ...(2)$$

Differentiate eqn. (1)

$$(dV_1) + (dV_2) = 0 \qquad \left\{ \begin{array}{l} dV_1 = (-Adx) \\ dV_2 = (Adx) \end{array} \right\}$$

$$(dV_1 = -dV_2)$$

$$W = \int (P_1 - P_2) Adx$$

$$= \int -P_1 dV_1 - \int P_2 dV_2$$

$$= (nRT) \int_{V_0}^{V_1} \frac{-dV_1}{V_1} - (nRT) \int_{V_0}^{V_2} \frac{dV_2}{V_2}$$

$$= -(nRT)\left[\ln\frac{V_1}{V_0} + \ln\frac{V_2}{V_0}\right]$$

$$= -(nRT)\left[\ln\frac{V_1 V_2}{V_0^2}\right]$$

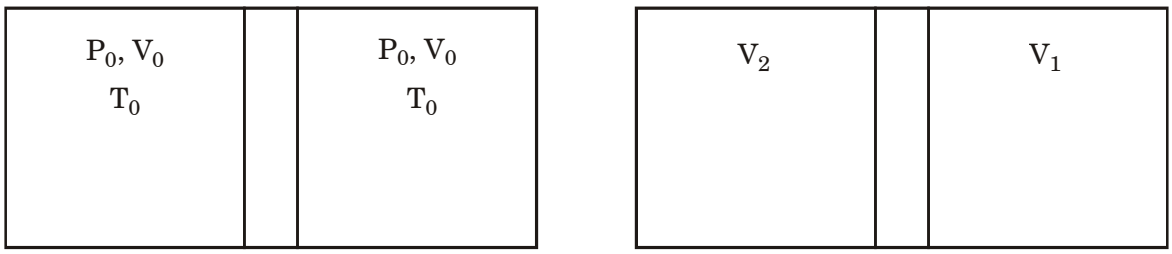
$$= P_0 V_0 \left[\ln\frac{V_0^2}{V_1 V_2}\right] = P_0 V_0 \left[\left(\frac{1+\eta}{2}\right)\left(\frac{1+\eta}{2\eta}\right)\right]$$

$$= P_0 V_0 \ln\left[\frac{(1+\eta)^2}{4\eta}\right]$$

Example 36

A thermally insulated horizontal cylinder is initially divided into two equal parts by a thin conducting piston which can freely move inside the cylinder. Each part contains, an ideal gas at temp. T_0. Find the work required to increase the volume of larger section η times the volume of the smaller section by slowly moving the piston.

Solution :



$$\frac{V_2}{V_1} = \eta$$

Temperature of both the chambers will be equal but changing

For the entire system, $dQ = 0$

$$dQ = dU + dW$$

$$dU + dW = 0$$

$$-dW = dU$$

$$\boxed{dW_{ex} + dW} = 0 \quad \rightarrow \quad \text{Work done by all force on piston will be zero}$$

$$dW_{ex} = -dW = dU$$

Now, $(nC_V dT)\ 2 = (P_1 - P_2)A\ dx$

$$= nRT\left(\frac{-dV_1}{V_1} - \frac{dV_2}{V_2}\right)$$

$$2 \times \left(\frac{nR}{\gamma - 1}\right) dT = -nRT\left(\frac{dV_1}{V_1} + \frac{dV_2}{V_2}\right)$$

$$\Rightarrow \quad \left(\frac{-2}{\gamma - 1}\right)\int \frac{dT}{T} = \int_{V_1}^{V_2} \frac{dV_1}{V_1} + \int_{V_0}^{V_2} \frac{dV}{V_2}$$

$$\left(\frac{2}{\gamma - 1}\right) \ln\left(\frac{T}{T_0}\right) = \ln\left(\frac{V_0^2}{V_1 V_2}\right) = \ln\left[\frac{(1+\eta)^2}{4\gamma}\right]^{\gamma - \frac{1}{2}}$$

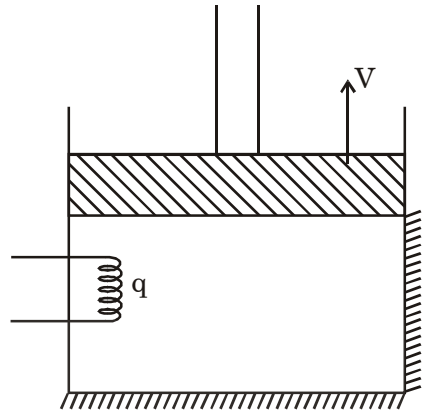
$$\left(\frac{T}{T_0}\right) = \frac{(1+\eta)^{\gamma-1}}{(4n)^{\gamma - 1/2}} \qquad \therefore \qquad \boxed{T = T_0 \frac{(1+\eta)^{\gamma-1}}{(4\eta)^{\frac{\gamma-1}{2}}}}$$

Work done by the system = $2nC_V dT$

Example 37

2 mol of an ideal monoatomic gas is enclosed in a vertical adiabatic cylinder fitted with an adiabatic piston of mass m, which can freely move inside the cylinder. Now a heater supplies a heat at rate of q J/s. Find the velocity of the piston under isobaric condition.

Solution :



$$\left(\frac{dq}{dt}\right) = nC_P\left(\frac{dT}{dt}\right) = 2 \times \left(\frac{5R}{2}\right)\left(\frac{dT}{dt}\right)$$

$$q = 5R\left(\frac{dT}{dt}\right)$$

$$dq = dU + dW$$

$$dq = nC_V dT + (PAdx)$$

$$n(C_P - C_V)\frac{dT}{dt} = \frac{PAdx}{dt}$$

$$\Rightarrow \quad 2 \quad R\ \frac{dT}{dt} = (PA)v$$

$$\left(\frac{2q}{5}\right) = PAv \quad \therefore \quad v = \left(\frac{2q}{5PA}\right)$$

$\therefore$ Velocity is constant, $\therefore$ net force on the piston is zero

$$PA = (P_0A + mg)$$

$$\therefore \quad \boxed{v = \frac{2q}{5(P_0A + mg)}}$$

HEAT TRANSFER

Heat can be transferred from one place to the other by any of three possible ways.

(1) Conduction : usually in solids

(2) Convection in fluids

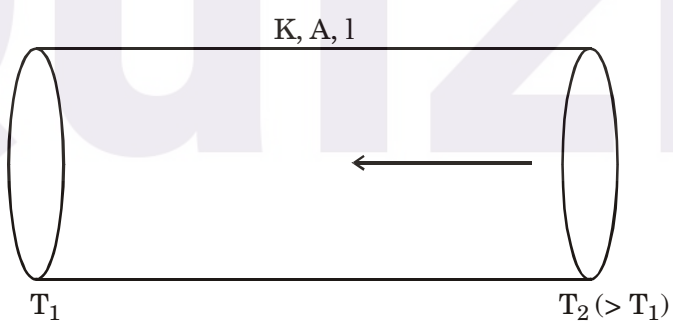
(3) Radiation : Vaccum, EM waves

In the first two processes a medium is necessary for heat transfer. Radiation does not require any medium and is the fastest mode of transfer.

CONDUCTION :

- Heat is transferred in conduction by collision with neighbouring particles within the body.
- When one end is placed at a higher temperature, atoms or molecules get heat energy and start oscillating with greater amplitude and it collides with neighbouring molecules and shares its KE. And in the process heat is transferred from one end to the other end.
- Average position of atoms/molecules do not change. (No actual transfers of mass is involved).
- In steady state, the heat crossing per unit time through any cross-section is constant. Which is given by,

$$\boxed{\frac{\Delta Q}{\Delta t} = -\frac{KA(T_2 - T_1)}{l}}$$



where K = thermal conductivity or conduction coefficient

Note :

(1) Area of cross-section should be taken perpendicular to direction of the heat flow.

(2) Length l should be measured along the direction of heat flow.

(3) In the steady state, the temperature of each cross section remains constant, (though it is not equal to all cross-section)

Limitations of $\frac{\Delta Q}{\Delta t} = \frac{KA(T_1 - T_2)}{l}$ formula

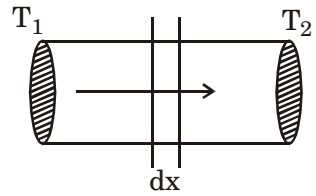
(i) To be used in the steady state only

(ii) Not applicable when area is not constant

(iii) Not applicable when K ≠ constant

However, in case of variable conditions, (eg. when K is a function of x) we may use :

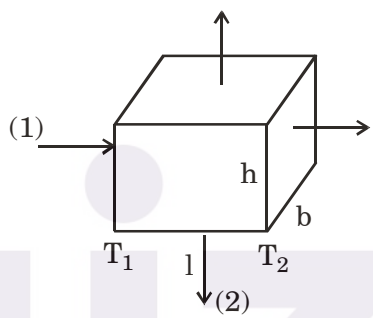
$$\frac{dQ}{dT} = -KA\frac{dT}{dx}$$



For example :

$$\left(\frac{\Delta Q}{\Delta T}\right)_1 = \frac{K(bh)(T_1 - T_2)}{l}$$

$$\left(\frac{\Delta Q}{\Delta T}\right)_2 = \frac{K(lb)\Delta T}{h}$$



Thermal Conductivity :

The ability of a material to conduct heat is measured in terms of thermal conductivity/conduction coefficient.

Greater the thermal conductivity, greater the tendency of the material to conduct heat

unit : W/m-K

Comparison with Electrical Quantities

$$i = \left(\frac{V_1 - V_2}{R}\right)$$

In heat conduction we have $\left(\frac{\Delta Q}{\Delta t}\right) = i = \frac{(T_1 - T_2)}{(d/KA)}$

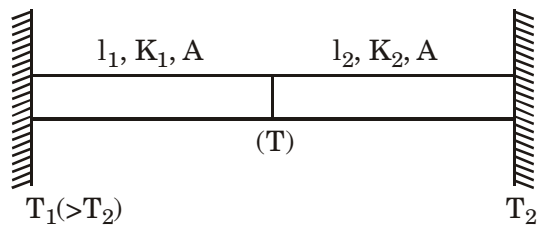
Potential difference ⇔ Temp. difference

Resistance ⇔ Thermal resistance = $\left(\frac{d}{KA}\right)$

Series and Parallel Combinations

(A) Series Combination :

(i) Thermal Resistance



$$\boxed{(Req) = R_1 + R_2}$$

Derivation :

$(T_1 - T) = iR_1$

$(T - T_2) = iR_2$

$(T_1 - T_2) = i(R_1 + R_2) = i(R_{eq})$

$\therefore$ $Req = (R_1 + R_2)$ $\left[R_1 = \frac{l_1}{K_1A}, \left(R_2 = \frac{l_2}{K_2A}\right)\right]$

(ii) Equivalent Thermal Conductivity in series combination

$l_{eq} = l_1 + l_2$

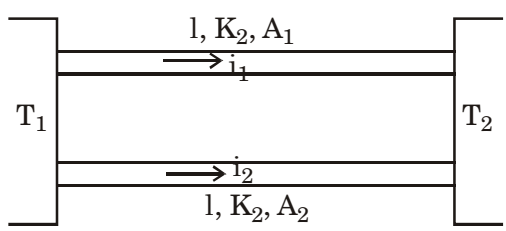
$$\Rightarrow \frac{(l_1 + l_2)}{KA} = \frac{l_1}{K_1A} + \frac{l_2}{K_2A}$$

$$\therefore \quad K = \frac{l_1 + l_2}{\left(\frac{l_1}{K_1} + \frac{l_2}{K_2}\right)} \qquad \therefore \quad K_{eq} = \frac{\sum_{i=1}^{n} l_i}{\sum_{i=1}^{n} \frac{l_i}{k_i}} = \frac{\int_0^l dx}{\int_0^l \frac{dx}{K}}$$

(B) Parallel Combination :

(i) Thermal resistance

$$R_1 = \frac{l}{K_1A_1}, R_2 = \frac{l}{K_2A_2}$$



$\therefore$ $i = (i_1 + i_2) \equiv$ Net heat flow / (time)

$$\Rightarrow \frac{(T_1 - T_2)}{R_{eq}} = \frac{(T_1 - T_2)}{R_1} + \frac{(T_1 - T_2)}{R_2}$$

$$\therefore \quad \boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots}$$

(ii) Equivalent Thermal Conductivity

$$\frac{KA}{l} = \frac{K_2A_1}{l} + \frac{K_2A_2}{l}$$

$$\therefore \quad K_{eq} = \left(\frac{K_1A_1 + K_2A_2}{A_1 + A_2}\right) = \frac{\sum K_iA_i}{\sum A_i} = \frac{\int KdA}{\int dA}$$

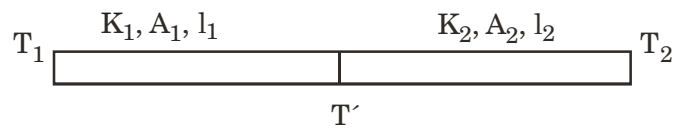
Temperature of Junction

To find the temp. of junction, first find the current i, and then use the resistance of the part upto which temperature is to be calculated.

e.g. $T_1 - T' = i\,R_1$

$T' = T_1 - i\,R_1$

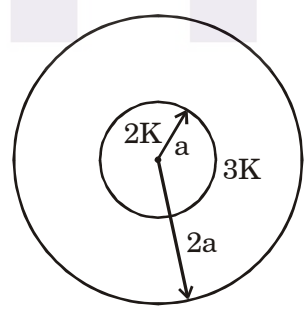
$$= T_1 - \frac{(T_2 - T_1)R_1}{\left(\frac{l_1}{K_1A} + \frac{l_2}{K_2A}\right)}$$



Example 38

Find the equivalent thermal conductivity of the given arrangement (of two concentric rods).

Solution :

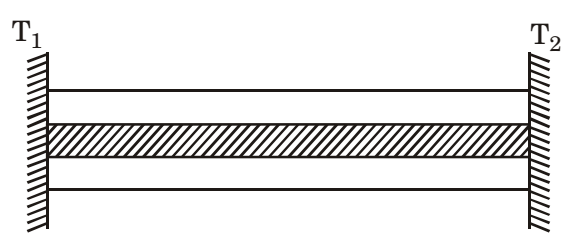


$$R_1 = \frac{l}{2K(\pi a^2)}$$

$$\therefore \quad R_2 = \frac{l}{9K(\pi a^2)}$$

Since the two rods are in parallel combinations

$$\therefore \quad R_{eq} = \frac{l}{11K(\pi a^2)} \quad \Rightarrow \quad \frac{l}{K'(4\pi a^2)} = \frac{l}{11K(\pi a^2)}$$



$$\therefore \quad K' = \left(\frac{11K}{4}\right)$$

Alternative (use formula)

$$\frac{KA}{l_1} = \frac{K_1A_1}{l_1} + \frac{K_2A_2}{l_2}$$

$$K = 2K\,\frac{\left(\pi a^2\right) + 3K\,\left(3\pi a^2\right)}{4\pi a^2} = \left(\frac{11K}{4}\right)$$

Example 39

Find the equivalent thermal conductivity when K varies as $K = K_0\left(1 + \frac{r}{a}\right)$ for the given rod.

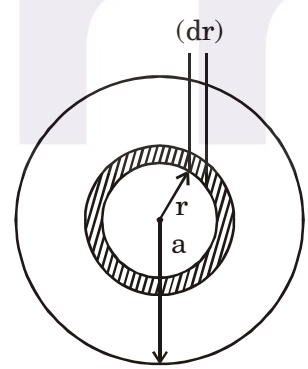
Solution :

$$K = K_0\left(1 + \frac{r}{a}\right)$$

Since all the small elements of thickness (dr) are is parallel combination

$(dA) = (2\pi r)dr$

$$K_{eq} = \frac{\int_0^a K\,dA}{\int_0^a dA} = \frac{\int_0^a K_0\left(1 + \frac{r}{a}\right)2\pi r\,dr}{\int_0^a (2\pi r)\,dr}$$



$$= K_0\,\frac{\left[\frac{a^2}{2} + \frac{a^2}{3}\right]}{\left[\frac{a^2}{2}\right]} = \left(\frac{5\,K_0}{6}\right)\times 2$$

$$\boxed{K_{eq} = \left(\frac{5\,K_0}{3}\right)}$$

Wheatstone Bridge

Consider the 5 rods AM, MB, AN, NB and MN connected as shown.

For balanced networks

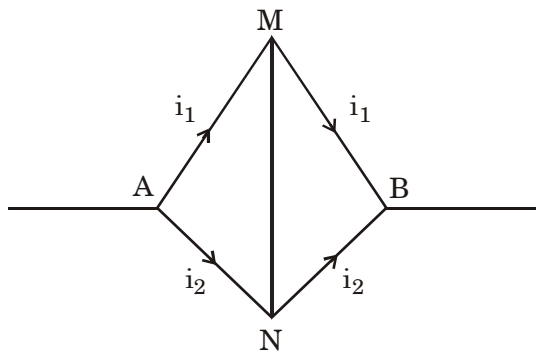
$$T_m = T_n$$

$$\Rightarrow \quad T_A - T_M = iR_1 \qquad T_M - T_B = i, R_3$$

$$T_A - T_N = i_2R_2 \qquad T_M - T_B = i_2R_4$$

$$\Rightarrow \quad i_1 R_1 = i_2R_2 \qquad i_1 R_3 = i_2R_4$$

$$\therefore \quad \left(\frac{R_1}{R_3} = \frac{R_2}{R_4}\right) \qquad \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

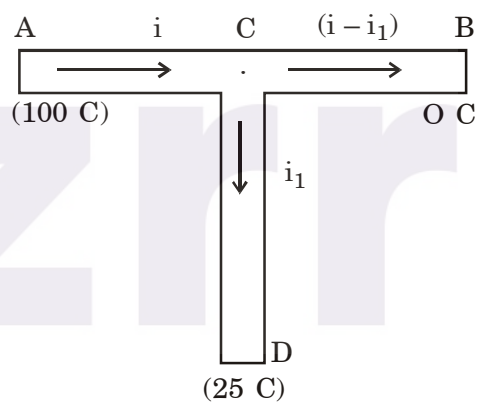


Junction law end loop are also applicable in thermal currents

Example 40

Two identical rods, each of R = 5 k/W, AB and CD are connected as shown.

Given T_A = 100 C, T_B = 0 C and T_D = 25 C. Find T_C.



Solution :

$$(T_A - T_C) = i(R_{AC})$$

$$(T_C - T_B) = (i - i_1)\ R_{CB}$$

$$(T_C - T_0) = i_1\ R_{CD}$$

$$\Rightarrow \quad (100 - T) = i(2.5)$$

$$(T - 0) = (i - i_1)\ (2.5)$$

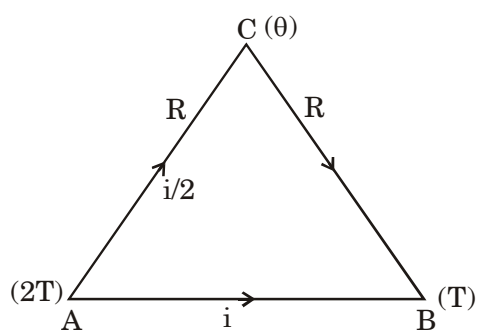
$$(T - 25) = i_1\ (5) \qquad \Rightarrow \qquad T_C = 45\ C$$

Example 41

3 identical rods made of same material form an equilateral triangle. Temperature of end A and Bare maintained at 2T and T respectively. Find the Temperature of the end C.

Solution :

AC and CB are in series and parallel on whole to AB

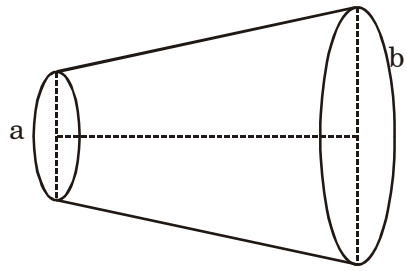


$$i_{AC} = \left(\frac{i_{AB}}{2}\right) \qquad (2T - T) = iR$$

$$(2T - \theta) = \left(\frac{i}{2}\right)R \qquad \left(\frac{2T - T}{2T - \theta}\right) = 2 \qquad \therefore \quad \theta = \left(\frac{3T}{2}\right)$$

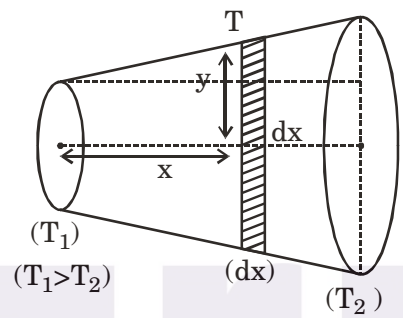
Example 42

Find the heat conducted per unit time along the length of the rod.



Solution :

Consider an infinitesimal element dx thickness at a distance x from one end



Use : $\left(\frac{\Delta Q}{\Delta t}\right) = i = -KA\left(\frac{dT}{dx}\right)$

$A = \pi y^2$

and, $\left(\frac{y-a}{x}\right) = \frac{(b-a)}{l}$

$\therefore \quad \left(\frac{l}{b-a}\right)(y-a) = x$

$dx = \left(\frac{l}{b-a}\right)dy \qquad \left(\frac{\Delta Q}{\Delta t}\right)\int\frac{dx}{y^2} = -K\pi\int dT$

$$\Rightarrow \quad \frac{\Delta Q}{\Delta t}\left[\left(\frac{l}{b-a}\right)\int_a^b\frac{dy}{y^2}\right] = -K\pi\int_{T_1}^{T_2} dT$$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta t}\right)\left(\frac{l}{b-a}\right)\left(\frac{-1}{b}+\frac{1}{a}\right) = K\pi(T_1 - T_2)$$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta t}\right)\left(\frac{l}{b-a}\right)\left(\frac{b-a}{ab}\right) = K\pi(T_1 - T_2)$$

$$\therefore \quad \left(\frac{\Delta Q}{\Delta t}\right) = \frac{K\pi(T_1 - T_2)ab}{l}$$

Thermal resistance $R = \left(\frac{l}{K\pi ab}\right)$

Alternative Method :

(Using thermal resistance)

Note : This alternative method cannot be used when the thermal conductivity K becomes function of temperature

$$\therefore \quad \int_0^R dR = \int_0^l \frac{dx}{K\pi y^2}$$

[All the thermal resistance are in series combination]

$$\therefore \quad R = \int_a^b \frac{l}{(b-a)} \frac{dy}{K\pi y^2} \qquad \left[x = \left(\frac{L}{b-a}\right)(y-a)\right]$$

$$= \frac{l}{(b-a)K\pi}\left(\frac{b-a}{ab}\right) = \frac{l}{K\pi ab}$$

$$\therefore \quad \frac{\Delta Q}{\Delta t} = \frac{(T_1 - T_2)}{(l/K\pi ab)} = \frac{K\pi ab(T_1 - T_2)}{l}$$

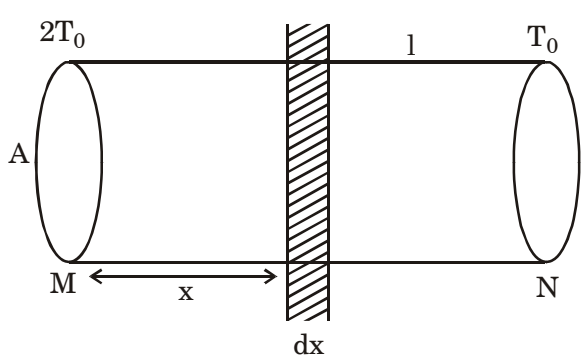
Learn : Thermal resistance of a frustum with radius of circular ends a and b = $\frac{l}{K\pi ab}$

Example 43

A uniform rod MN of length l and cross-sectional area A with thermal conductivity given by $K = \frac{K_0}{(1+2T)x}$ has its two ends at temperature $2T_0$ and T_0. Find

(a) Heat conducted per unit time through the rod.

(b) The temperature of the rod as a function of distance x.



Solution :

(a) Heat current $i = -KA\left(\frac{dT}{dx}\right)$

$$\left(\frac{\Delta Q}{\Delta t}\right) = \frac{-K_0\ A}{(1+2T)x}\left(\frac{dT}{dx}\right)$$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta t}\right)\int_0^l x\,dx = -K_0A\int_{2T_0}^{T_0}\frac{dT}{(1+2T)}$$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta t}\right)\left(\frac{l^2}{2}\right) = \frac{-K_0A}{2}\ln\left(\frac{1+4T_0}{(1+2T_0)}\right)$$

$$\left(\frac{\Delta Q}{\Delta t}\right) = \frac{2K_0A}{2l^2}\ln\left(\frac{1+4T_0}{(1+2T_0)}\right)$$

(b) Temperature of a rod as a function of x can be calculated using.

$$\left(\frac{\Delta Q}{\Delta t}\right) = \int_0^x x dx = -\int_{2T_0}^{T}\frac{KAdT}{(1+2T)}$$

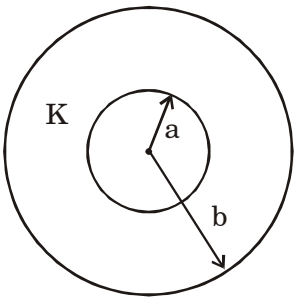
$$\left(\frac{\Delta Q}{\Delta t}\right)\left(\frac{x^2}{2}\right) = K_0A\ln\left(\frac{1+4T_0}{1+2T}\right)$$

$$\left[\frac{2K_0A}{l^2}\ln\left(\frac{1+4T_0}{1+2T_0}\right)\right]\frac{x^2}{2} = K_0A\ln\left(\frac{1+4T_0}{1+2T}\right)$$

$$\Rightarrow \quad \left(\frac{x^2}{l^2}\right)\ln\left(\frac{1+4T_0}{1+2T_0}\right) = \ln\left(\frac{1+4T_0}{1+2T}\right)$$

$$\boxed{x^2 = l^2\frac{\ln\left(\frac{1+4T_0}{1+2T}\right)}{\ln\left(\frac{1+4T_0}{1+2T_0}\right)}}$$

Example 44



Find the heat conducted per unit time when

(a) K = const

(b) $K = (\alpha/_r)$

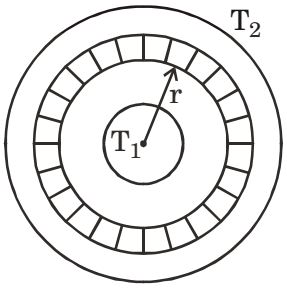
(r ≡ distance from centre)

Solution :

(a) Heat conduction through a spherical shell

$$\left(\frac{\Delta Q}{\Delta T}\right) = -KA\left(\frac{dT}{dr}\right)$$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta T}\right) = -K\left(4\pi r^2\right)\frac{dT}{dr}$$



$(T_1 > T_2)$

$$\Rightarrow \quad \left(\frac{\Delta Q}{\Delta T}\right)\int_a^b \frac{dr}{4\pi r^2} = -K\int_{T_1}^{T_2} dT$$

$$\Rightarrow \quad \frac{\Delta Q}{\Delta T}\left|\frac{b-a}{ab}\right|\frac{1}{4\pi} = -K\left(T_2 - T_1\right)$$

$$\therefore \quad \frac{\Delta Q}{\Delta T} = \frac{4\pi Kab}{(b-a)}\left(T_1 - T_2\right)$$

Alternative Method

$$R = \int_a^b \frac{dr}{K\left(4\pi r^2\right)}$$

$$R = \frac{1}{4\pi K}\left(\frac{b-a}{ab}\right)$$

$$\therefore \quad \frac{\Delta Q}{\Delta t} = \frac{\left(T_1 - T_2\right)}{R} = \frac{\left(T_1 - T_2\right)4\pi Kab}{(b-a)}$$

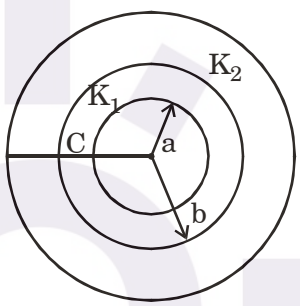
Learn : Thermal resistance of the spherical shell = $\dfrac{b-a}{4\pi Kab}$

(b) When K varies as $K = \frac{\alpha}{\gamma}$

$$R = \int \frac{dr}{KA}$$

$$= \int_a^b \frac{dr}{\left(\frac{\alpha}{r}\right)(4\pi r)} = \frac{1}{(4\pi r)} \ln\left(\frac{b}{a}\right)$$

Series Combination of Spherical Shell



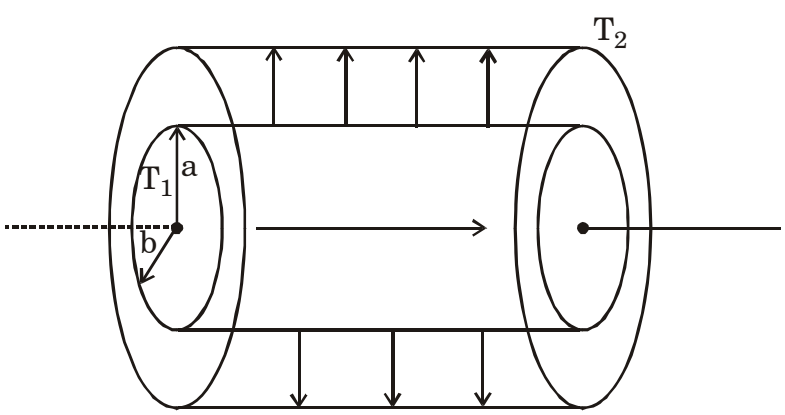
$$R_1 = \frac{(b-a)}{4\pi Kab}$$

$$R_0 \quad \frac{(c-b)}{(4\pi Kbc)}$$

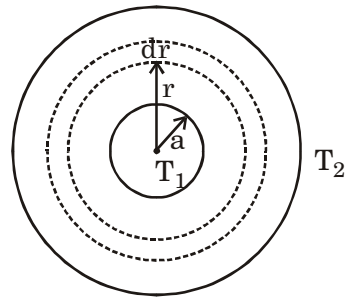
$$\therefore \quad R_{eq} = R_1 + R_2$$

$$\left(\frac{\Delta Q}{\Delta t}\right) = \frac{(T_1 - T_2)}{(R_{eq})} = i$$

Heat Conduction through cylindrical shell



(a) $\left(\frac{\Delta Q}{\Delta t}\right) = -K(2\pi rL)\frac{dT}{dr}$



$$\left(\frac{\Delta Q}{\Delta t}\right)\int_a^b \frac{dr}{2\pi rL} = -K\int_{T_1}^{T_2} dT$$

$$\left(\frac{\Delta Q}{\Delta t}\right) = \frac{K(T_1 - T_2)\,2\pi L}{\ln\left({}^{b}/_{a}\right)}$$

Alternative Method :

$$R = \int_a^b \frac{dr}{K\,(2\pi rL)} \qquad \therefore \quad \frac{\Delta Q}{\Delta t} = \left(\frac{T_1 - T_2}{R_{eq}}\right)$$

$$= \frac{\ln\left({}^{b}/_{a}\right)}{(2\pi K)L}$$

(b) When $K = ({}^{\alpha}/_{r})$

$$R = \int_a^b \frac{dr}{\frac{a}{r}(2\pi rL)} = \frac{(b-a)}{2\pi\alpha L}$$

$$\therefore \quad \left(\frac{\Delta Q}{\Delta t}\right) = \frac{(2\pi\alpha L)}{(b-a)}(T_1 - T_2)$$

Example 45

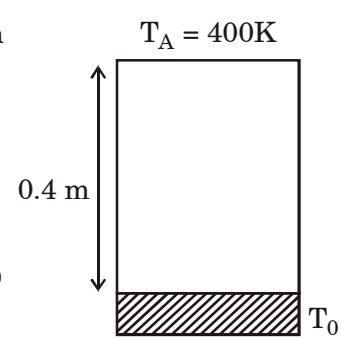
A cylinder of length 0.4 m and area A = 0.04 m^2 is placed on a thin circular disc of same cross-sectional area.

T_A = 400 K and T_0 = 200 K

T_A is maintained constant

K = 10 W/M, S_d = 600 J/g/K, m_{disc} = 0.4 kg

Find the time when the temperature of disc becomes (T = 350 K)



Solution :

At any time instant let, temperature of the disc be T. We have

$$\left(\frac{dQ}{dt}\right) = \frac{KA\left(T_A - T\right)}{l}$$

$$\Rightarrow \quad mS\frac{dT}{dt} = \frac{KA\left(T_A - T\right)}{l}$$

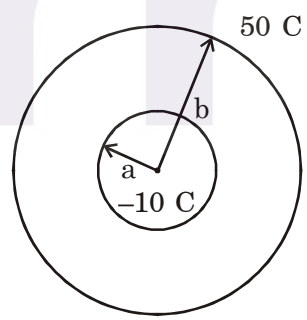
$$\Rightarrow \quad \int_{300}^{350} \frac{dT}{\left(T_A - T\right)} = \frac{KA}{mSl}\int_0^t dt$$

$$\Rightarrow \quad -\left|\ln\left(T_A - T\right)\right|_{300}^{350} = \left(\frac{KA}{mSl}\right)t$$

$$\therefore \quad t = \left(\frac{mSl}{KA}\right)\ln\left(100/50\right)$$

$$t = \frac{0.4 \times 600 \times 0.4}{10 \times 0.04} \times \left(\ln 2\right)$$

Example 46



An ice ball of mass m and radius a is placed inside a spherical conducting shell of radius (a and b) at – 10 C. The thermal conductivity varies as $K = \left(\frac{\alpha}{r}\right)$ of the spherical shell. Find the time required to increase the temperature of ice – 10 C to + 10 C.

Solution :

First, find the thermal resistance of the spherical shell.

$$R = \int_a^b \frac{dr}{K\left(4\pi r^2\right)} = \int_a^b \frac{dr}{\left(\frac{\alpha}{r}\right)\left(4\pi r^2\right)} = \frac{1}{4\pi\alpha}\left(\ln b/a\right)$$

$$\therefore \quad \left(\frac{\Delta Q}{\Delta t}\right) = \frac{\left(T_1 - T_2\right)}{\left(R_{eq}\right)}$$

Now divide the time into three intervals

(i) t_1 : time taken to increase temperature from – 10 C to 0 C

$$m(S_{ice})\left(\frac{d\theta}{dt}\right) = \frac{(50-\theta)}{R}$$

$$\int_{-10°C}^{0°C} \frac{d\theta}{(50-\theta)} = \int_0^{t_i} \frac{dt}{(mS_iR)}$$

$$-\ln\left(\frac{50}{60}\right) = \frac{t}{(mS_i\ R)}$$

$$\therefore \quad t_1 = \left[\frac{mS_i\ \ln(b/a)}{4\pi\alpha}\right] \ln(6/5)$$

(ii) t_2 : time required to melt the ice from 0 C to 0 C water

$$\left(\frac{mL_f}{dt}\right) = \frac{(50-0)}{R}$$

$$\therefore \quad t_2 = \frac{(mL_f)}{50} = \frac{(m \times L_f)\ \ln(b/a)}{(4\pi a)\ 50}$$

(iii) t_3 : time required to increase the temperature from 0 C to 10 C water

$$mS_W\left(\frac{d\theta}{dt}\right) = \frac{50-\theta}{R}$$

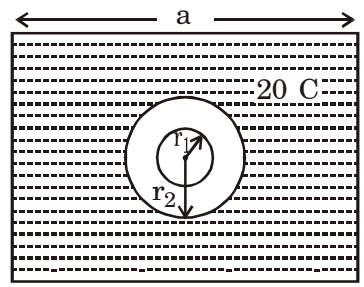
$$\therefore \quad \int_0^{10} \frac{d\theta}{(50-\theta)} = \int_0^{t_3} \frac{dt}{ms_WR}$$

$$\therefore \quad t_3 = (ms_WR)\ \ln(5/4)$$

$\therefore$ total time = $(t_1 + t_2 + t_3)$ (required)

Example 47

A cubical box is filled with water. Now a hot sphere of radius r_1 kept at const. temp. of 1000 C has a jacket around it of radius ($r_2 = er_1$) to jacket consist of this concentric spherical layers of $K = \left(\frac{C}{r}\right)$. Find the time required to increase the temperature from 20 C to 30 C.



Given that :

$$\ln\left(\frac{98}{97}\right)\left[a^3 - \frac{4}{3}\pi r_2^3\right]\rho_W S_W = 400\pi C$$

Solution :

Thermal resistance of the jacket :

$$R = \int_{r_1}^{r_2} \frac{dr}{K\left(4\pi r^2\right)} = \int_{r_1}^{r_2} \frac{dr}{\left(\frac{C}{r}\right)4\pi r^2} = \frac{1}{4\pi}\ln\left(r_2/r_1\right) = \frac{\ln e}{4\pi C}$$

$$\therefore \quad R = \left(\frac{1}{4\pi C}\right)$$

Now, consider the heat current flowing from the sphere to water in the box

$$ms_W\left(\frac{d\theta}{dt}\right) = \frac{(1000-\theta)}{R}$$

$$\int_{20°C}^{30°C} \frac{d\theta}{(1000-\theta)} = \left[\int_0^t dt\right]\frac{1}{ms_W K}$$

$$-\ln\left(\frac{97}{98}\right) = \left[\frac{1}{(ms_W R)}\right]t$$

$$\Rightarrow \quad -\ln\left(\frac{98}{97}\right)\rho_W\left[a^3 - \frac{4}{3}\pi r_2^3\right]S_W = (4\pi C)t$$

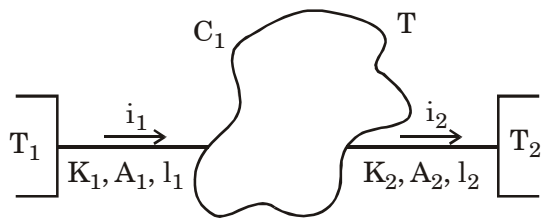
$$\Rightarrow \quad 400\pi C = 4\pi C t$$

$$\therefore \quad \boxed{t = 100\text{ s}}$$

Example 48

A body of heat capacity C is connected with two rods whose free ends are kept at constant temp of T_1 and T_2. Let the initial temp. of the body be T_0 where ($T_1 > T_0 > T_2$). Find the temperature of the body as a function of time t.

Solution :



Let the temp. at any instant be T.

Concept : Heat trap by the body = $(i_1 - i_2)$ (which increases the temp. of the body)

$$i = (i_1 - i_2)$$

$$ms\left(\frac{dT}{dt}\right) = \frac{K_1 A_1 (T_1 - T_0)}{l_1} - \frac{K_2 A_2 (T_0 - T_2)}{l_2}$$

$$ms\frac{dT}{dt} = \left(\frac{K_1\ A_1\ T_1}{l_1} + \frac{K_2\ A_2\ T_2}{l_2}\right) - \left(\frac{K_1\ A_1}{l_1} + \frac{K_2\ A_2}{l_2}\right) T_0$$

$$\left(\frac{1}{C}\right)\frac{CdT}{dt} = \left(\frac{K_1 A_1\ T_1}{l_1} + \frac{K_2\ A_2\ T_2}{l_2}\right)\frac{1}{C} - \left(\frac{K_1\ A_1}{l_1} + \frac{K_2\ A_2}{l_2}\right)\frac{1}{C}T$$

$$\frac{dT}{dt} = (a - bT)$$

$$\int_{T_0}^{T}\left(\frac{dT}{a - bT}\right) = \int_0^t dt$$

$$\ln\frac{(a - bT)}{(a - bT_0)} = -bt$$

$$(a - bT) = (a - b\ T_0)\ e^{-bt}$$

$$\boxed{\therefore \quad \frac{a - (a - b\,T_0)e^{-0bt}}{b} = T}$$

CONVECTION

Takes place in fluids

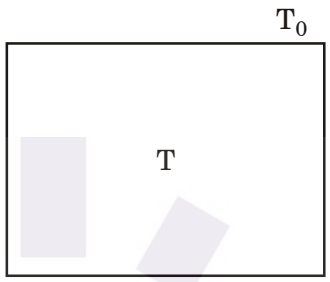
Convection is of following 2 types :

(a) **Natural Convections :**

In convections, heat transfers takes place due to motion of heated material and in natural convection this transfer is due to difference in density.

(b) **Forced Convection :**

If the heated material is forced to move by any external source such as pump, fan etc., then such a convection is called forced convection.



Heat transfer through convection per unit time

$$\left(\frac{dQ}{dt}\right) = hA\left(T - T_0\right)$$

where

A ≡ surface area of the body is contact with the fluids.

h ≡ convection co-efficient which depend on the nature of the fluids in contact with the surface.

RADIATION

– Radiation involves heat transfer through EM waves

– Medium is not required

– At all temp., all bodies continuously emit heat energy through radiations.

(1) Emissive power (E) : The radiant energy emitted per unit time per unit surface area of the body is called emissive power and is denoted by E.

$$E = \frac{\Delta U}{A\left(\Delta t\right)}$$

units : W/m^2

– The radiant energy is emitted in all wave length range from 0 to ∞, however, only a small range corresponds to larger part of the radiant energy.

(2) Spectral Emissive Power (E_λ) : The radiant energy emitted per unit time, per unit area per unit wavelength is called spectral emissive power.

– In the other words, the emissive power for a particular wavelength is called spectral emissive power (E_λ).

$$E_\lambda = \left(\frac{dE}{d\lambda}\right)$$

or $E = \int_0^\infty (E_\lambda \, d\lambda) \equiv$ total emissive power

(3) Absorptive power (a) :

The fraction of incident radiant energy absorbed by the surface of the body is called absorptive power.

$$a = \frac{\text{absorbed incident radiant energy}}{\text{total incident radiant energy}}$$

for a perfect black body, $a = 1$

for a perfect white body, $a = 0$

for a grey body, $0 < a < 1$

Note : The incident radiant energy on the surface of the body is divided into 3 parts :

(i) Absorbed part, (ii) Transmitted part (iii) Reflected

Also, $\alpha + \beta + \gamma = 1$ for any body having incident radiant energy where $\alpha \equiv$ absorptivity, $\beta \equiv$ transmitivity, $\gamma \equiv$ reflectivity

(4) Spectral Absorptive Power (a_λ)

The absorptive power for a particular wavelength is called spectral absorptive power.

$$a_\lambda = \left(\frac{da}{d\lambda}\right)$$

$\Rightarrow \quad a = \int_0^\infty a_\lambda \, d\lambda \equiv$ total absorptive power

$a \equiv$ area under the $a_\lambda - \lambda$ curve

(5) Kirchoff's Law : It states that

(i) At a given temperature, the ratio of emissive power to the absorptive power is same for all bodies.

(ii) and is equal to the emissive power of the black body at that temperature.

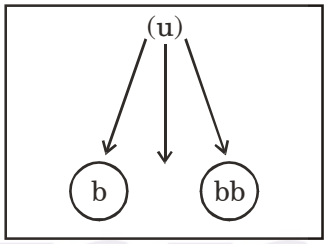
$$\left(\frac{E_b}{a_b}\right) = (E_{bb}) \text{ or } \left(\frac{E_b}{E_{bb}}\right) = a_b$$

Also, $\frac{(E_b)_\lambda}{(a_b)_\lambda} = (E_{bb})_\lambda$

In conclusion : "A good emitter is a good absorber and vice-versa".

Proof :

Consider two bodies, one grey and the other a perfect black body, enclosed in an isolated container, which have identical dimension and are in thermal equilibrium



In thermal equilibrium,

Amount of radiant energy falling on black body is completely reflected black ($\because$ its temperature is constant)

$\therefore \quad E_{bb} = \frac{U}{A(\Delta t)}$, $E_b = \frac{a_b U}{A(\Delta t)}$ [Energy emitted by body its temperature is constant]

$\therefore \quad E_b = a_b E_{bb}$

$$\left(\frac{E_b}{a_b}\right) = E_{bb}$$

(6) On the base of quantum theory of radiation, plank has shown that the energy distribution in the spectrum of black body radiation is given by : $R_\lambda = \frac{2\pi hc^2}{\lambda^5}\left[\frac{1}{e^{ch/\lambda KT} - 1}\right]$

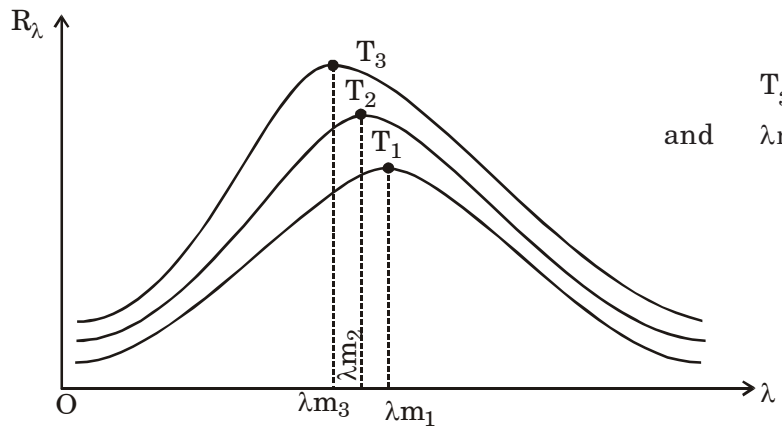
where $R_\lambda \equiv$ spectral radiancy or spectral intensity for a given wavelength.

(Radiancy is the intensity due to radiation)

$T \equiv$ absolute temp; $h \equiv$ planck's constant

$K \equiv$ Boltzmann const = 1.38 10^{23} J/k

$C \equiv$ speed of light



$T_3 > T_2 > T_1$

and $\lambda m_3 < \lambda m_2 < \lambda m_1$

(i) The area under the graph curve, gives the total intensity of radiation, (radiancy) at a particular temperature.

$$R = \int_0^\infty R_\lambda \, d\lambda$$

And, this area increases with the increase in temperature.

(ii) Radiancy is found to be proportional to the fourth power of absolute temperature of the surface of the body. (Stefan's law)

$R \propto T^4$

$R = \sigma T^4$

$U = (\sigma A T^4)$

(7) Wein's Law : The wavelength for maximum intensity of radiation is inversely proportional to the temp. of the surface

$$\boxed{\lambda_m \propto \frac{1}{T}}$$

$\Rightarrow$ $\lambda_m T = b$ (Wein's constant)

$b = 2.88 \quad 10^{-3}$ m–K

Wein's law can be used to determine the temperature of any surface emitting the radiation.

(8) Stefan's Law : The radiant energy emitted per unit time by the surface of the body is given by

$$u = e \sigma A T^4$$

$A \equiv$ total surface area of the body exposed to the surrounding

$T \equiv$ absolute temp. of the surface of the body

$e \equiv$ emissivity of the surface

σ = Stefan's constant

$= 5.67 \quad 10^{-8}\ W/m^2 - K^4$

Note :

(1) Emissivity of a black body = 1

Also, emissivity of any body = absorptive power of the body

$$\frac{E_b}{E_{bb}} = (a_b)$$

$$\Rightarrow \frac{E(\sigma AT^4)}{(\sigma AT^4)} = a_b \qquad \Rightarrow \qquad \boxed{E = (a_b)}$$

(2) The heat energy lost per unit time by a body t_0 is given by :

$$\boxed{\Delta u = e\sigma A\ (T^4 - T_0^{\ 4})}$$

where $T \equiv$ temp. of the body

$T_0 \equiv$ temp. of the surrounding

$A \equiv$ Area of the body exposed to the surrounding

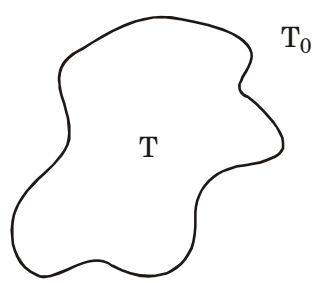
$(T > T_0)$

Similarly heat gained per unit time from the surrounding will be given by $= e\sigma A\ (T^4 - T_0^{\ 4})$

Newton's Cooling Law

$\Delta T = (T - T_0)$

$\Delta T << T_0$ (in K)



Heat const. per unit time due to convection,

$\Delta u_1 = hA\ (T - T_0)$

Heat lost per unit time due to radiation

$\Delta u_2 = e\sigma A\ (T^4 - T_0^{\ 4})$

$$= e\sigma A\ T_0^4 \left[\left(1 + \frac{\Delta T}{T_0}\right)^4 - 1\right]$$

$\therefore \quad \Delta T << T_0$

$$= e\sigma A\ T_0^4\left[1+\frac{4\Delta T}{T_0}-1\right]$$

$$= \left(4e\sigma AT_0^3\right)\left(\Delta T\right)$$

Total heat loss per unit time

$$\Delta u = \Delta u_1 + \Delta u_2$$
$$= (h + 4e\sigma AT_0{}^3)A\ (T - T_0)$$

$$-ms\left(\frac{dT}{dt}\right) = \left(h + 4e\sigma AT_0^3\right)A\left(T - T_0\right)$$

$$\therefore \quad -\left(\frac{dT}{dt}\right) = \left(\frac{h + 4e\sigma AT_0^3}{ms}\right)A\left(T - T_0\right)$$

$$-\left(\frac{dT}{dt}\right) = K\left(T - T_0\right) \text{ (in Kelvin)}$$

K $\equiv$ cooling constant

$$= \left(\frac{h + 4e\sigma T_0^3}{ms}\right)A$$

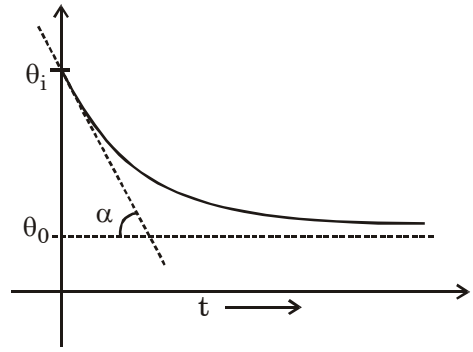
$$\boxed{\left(\frac{d\theta}{dt}\right) = K\left(\theta - \theta_0\right)} \text{ in celsius}$$

$$\int_{\theta i}^{\theta}\frac{-d\theta}{\theta - \theta_0} = K\int_0^t dt$$

$$\frac{\ln\left(\theta - \theta_0\right)}{\left(\theta_i - \theta_0\right)} = -Kt$$

$$\boxed{\theta = \theta_0 + \left(\theta_i - \theta_0\right)e^{-Kt}}$$

Cooling curve



$$\left(\frac{d\theta}{dt}\right) = -\tan\alpha$$

$$= -K(\theta - \theta_0)$$

$\therefore$ $\tan\alpha = K(\theta - \theta_0)$

$\therefore$ $K = \dfrac{\tan\alpha}{(\theta - \theta_0)}$

$\therefore$ to calculate the value of K, use the tangent of the slope at point and the temp. at that instant.

Example 49

A body initially at temp. θ_1 is kept in a surrounding of temp. θ_0 then find :

(i) the maximum heat that can be lost by the body

(ii) the time in which 90% of the heat will be lost by body. (m = mass, S_1 ≡ specific heat)

Solution :

(i) $\Delta Q_{max} = ms(\theta_1 - \theta_0)$

(ii) $\Delta Q = ms(\theta_1 - \theta_2)$

$\Delta Q = 0.9\ \Delta Q_{max}$

$ms(\theta_1 - \theta_2) = (0.9)\ ms(\theta_1 - \theta_0)$

$\theta_1 - \theta_2 = 0.9\ \theta_2 - 0.9\ \theta_0$

$(0.1\ \theta_1 + 0.9\ \theta_0) = \theta_2$

$\therefore$ $\dfrac{-d\theta}{dt} = K(\theta - \theta_0)$

$$\int_{\theta_1}^{\theta_2} \frac{-d\theta}{(\theta - \theta_0)} = K\int_0^t dt$$

$$\left|\ln\frac{(\theta_2 - \theta_0)}{(\theta_1 - \theta_0)}\right| = -Kt$$

$\ln(0.1) = -Kt$

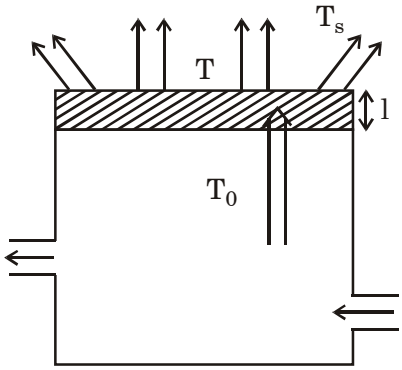
$Kt = \ln 10$

$\Rightarrow$ $\boxed{t = \dfrac{2.303}{K}}$

General concept involved in questions based on conduction and radiation.

In most of the questions it is to be used that the heat conductive per unit time due to conduction is equal to loss through radiation.

eg.



In steady state,

$$\frac{KA(T_0 - T)}{l} = Ae\sigma\left(T^4 - T_s^4\right)$$